MDPI

Article

WRF Cold-Pool Structure and Evolution during a Uinta Basin Winter Ozone Episode

John R. Lawson [1,2,*], Michael J. Davies [1], Loknath Dhar [1,3] and Seth N. Lyman [1,3]

[1] Bingham Research Center, Utah State University, Vernal, UT 84078, USA;
[2] Department of Mathematics and Statistics, Utah State University, Logan, UT 84322, USA;
[3] Department of Chemistry and Biochemistry, Utah State University, Logan, UT 84322, USA
* Correspondence: john.lawson@usu.edu

Abstract

Cold-air-pool structure sets the mixing volume and transport wind that a winter-ozone chemistry model inherits from its meteorological driver, and surface skill alone does not establish it. We compared four deterministic Weather Research and Forecasting (WRF) simulations of the 2 February 2013 Uinta Basin episode, crossing GFS and NAM driving products with one-way and two-way nest feedback at eleven fixed 1 km station columns over five post-spin-up hours (14:00–18:00 UTC). Relative to NAM, GFS increased the paired 2 m-temperature-bias contrast by 1.79 and 1.76 °C under two-way and one-way feedback, reduced 50–500 m stability by 3.63 and 3.65 K, and increased fixed-layer transport wind by 0.79 and 0.99 m s^{-1}, with the same sign in all five hours. Over a registered evaluation footprint, the GFS members also carried 0.70 to 1.80 MJ m^{-2} less integrated heat deficit than the feedback-matched NAM members at every retained hour under seven footprint definitions. Heat deficit nevertheless increased in every member, so a predeclared test for faster GFS heat-deficit loss failed in all fourteen pair-definition combinations. Surface-pressure errors ordered the products oppositely, and within-product feedback effects were much smaller throughout. A matched recomputation of the feedback contrast at the fine-terrain control pair reproduced that bound: every fine-terrain feedback effect was one to two orders of magnitude below the matched driving-product contrast. This working revision additionally surveys the archived controlled comparisons that surround the factorial—spatial and vertical response structure, terrain-source brackets, slope-aware radiation, boundary-layer treatment, and initialization context—with each element labeled by its evidence status rather than withheld. All five evidence gates of the selected results architecture have now been executed with predeclared criteria: the fine-terrain feedback bound passed; the spatial-coherence and terrain gates failed narrowly defined criteria and are reported as negatives while their confinement and lineage legs passed; and a coherent boundary-layer-scheme response was selected as the one remaining process candidate. The experiment does not separate initial from lateral-boundary forcing, validate inversion depth, establish multi-day persistence, or predict ozone.

Keywords: winter ozone; WRF; cold-air pool; Uinta Basin; driving data; nest feedback; heat deficit; model evaluation

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Air* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

[Diagnostic — not gate-passed: Working-draft mode. This revision is deliberately over-inclusive: it carries every archived controlled comparison, candidate figure, and

mechanism hypothesis that the study's evidence records can currently support at any status, so that coauthors can prune from a complete display rather than request missing material. Three status classes appear throughout. *Promoted* figures and *confirmed* claims have passed the repository's provenance and admission gates and carry ledger identifiers (C-numbers). *Registered candidates* are tracked with full provenance but are not admitted. *Exploratory renders* are batch engine figures catalogued in `figures-draft/HARVEST.md`; they illustrate visual leads only, and no quantitative claim in this draft rests on them. The analysis gates named in this draft (ANA-17, ANA-18, ANA-19, ANA-20) have all been executed with criteria fixed in committed scripts before any output was read; each section reports its gate's pass or fail outcome, and the complete decision record is `rewrite/HYBRID-GATE-REVIEW.md`. A submission-ready revision would remove most exploratory material or promote registered candidates through the normal admission path.]

Winter ozone in the Uinta Basin develops when photochemistry proceeds within snow-covered, stably stratified air that restricts vertical dilution and horizontal exchange [1–3]. Snow increases actinic flux and modifies deposition, while the stable, weakly ventilated air confines ozone and its precursors near the basin floor. For a coupled model, the meteorological question is therefore not only whether 2 m temperature has a small mean error. It is whether the atmosphere delivered to the chemistry grid contains a credible shallow mixing environment and weak transport during the accumulation period.

Published concentration agreement does not provide a clean ranking of the relevant uncertainties. Ahmadov et al. obtained a much stronger ozone simulation after applying an aircraft-derived, Horsepool-centered emissions scaling while holding the meteorological simulation fixed, and they conditioned the resulting inventory interpretation on adequate meteorology [1]. Matichuk et al. evaluated a cumulative ladder that changed volatile-organic-compound emissions, leaf area, and deposition together, while improved WRF temperature profiles did not consistently improve ozone profiles [4]. These outcomes are consistent with compensating error among confinement, emissions, photolysis, deposition, and chemical mechanism. They make meteorological evaluation necessary for chemical attribution without making meteorology the uniquely dominant error source.

The meteorological literature also warns against reducing an experiment to a configuration label. Neemann et al. showed strong case-specific responses from bundled snow, albedo, and cloud-microphysics choices, but their ice-retention modification was a bounding treatment and the 1.33 km grid evaluated most closely was finer than the 4 km meteorology supplied to the chemistry model [2]. Tran et al. showed that the meteorological response to four-dimensional data assimilation depended on the nudged variables, input data, and diagnosed boundary-layer depth; their disclosed artificial formaldehyde limits quantitative chemical inference [5]. Thus, labels such as "NAM driven," "nudged," or "high resolution" do not by themselves identify a controlled cause, and aggregate scores can obscure a failure to retain the layer of interest.

The present study asks three deliberately bounded questions. First, does the short-window lower-layer state differ between GFS- and NAM-driven WRF simulations under both nest-feedback modes? Second, how large is the within-product nest-feedback response relative to that driving-product contrast? Third, does the driving-product response also appear in a footprint-integrated heat-deficit measure of the simulated cold pool, and do the products differ in how that measure changes across the window? We answer these questions with paired surface errors, common-height stability and wind diagnostics, and an area-weighted heat-deficit integral on one domain, station field, and five-hour window. The heat-deficit comparison carries two criteria that were fixed before any output was inspected, and we report the outcome of both. The design evaluates the meteorological

input state that a later chemistry calculation would receive; it does not contain a chemistry calculation.

Beyond those three anchored questions, the archive built for this study contains a larger family of controlled comparisons: the same driving-product contrast under a second terrain regime, a replication of the feedback bound at the fine-terrain control, terrain-source brackets on a fixed grid, a slope-aware radiation pair, a boundary-layer and surface-layer scheme pair, and vertical and upper-air structure diagnostics. This working revision presents that family in full, with each comparison labeled by design status, analysis status, and figure status, so that the eventual submission can select from evidence rather than from recollection. The fine-terrain feedback recomputation is complete and passed its gate (Section 3.8); the spatial-coherence and terrain gates were executed and failed predeclared criteria, and their outcomes are reported quantitatively as negatives (Sections 3.5 and 3.9); the process screen selected the boundary-layer-scheme comparison as the one remaining candidate result (Section 3.10).

2. Materials and Methods

2.1. Factorial Design and Estimands

The experiment crossed the archived GFS analysis product `gfsanl_4` and NAM analysis product `namanl_218` with one-way (`feedback=0`) and two-way (`feedback=1`) nest feedback (Table 1). Stable study codes connect each member to the run and comparison inventories. The products supplied zero-hour analyses at 12:00 and 18:00 UTC and provided 27 and 40 metgrid levels for GFS and NAM, respectively. The GFS-versus-NAM factor is therefore a product-level treatment: it bundles the products' native grids, analysis systems, vertical input, assimilated observations, 12:00 UTC initial state, and subsequent lateral-boundary evolution. The experiment cannot assign the resulting response to one of those components. Within each product family, the normalized as-run namelists differed only in the feedback flag. The two product contrasts were evaluated separately so that the driving-product response had to recur under both feedback settings to support the primary result. The two within-product feedback contrasts and their paired interaction bounded the secondary response.

Table 1. Driving-product by nest-feedback experiment matrix. The study codes resolve to exact run paths and fingerprints in the run inventory.

Study code	Driving-data product	Nest feedback	Manuscript label
X6	GFS	two-way	GFS two-way
X10	GFS	one-way	GFS one-way
X5	NAM	two-way	NAM two-way
X3	NAM	one-way	NAM one-way

2.2. Case, Observations, and Common Sample

The target is the snow-covered Uinta Basin cold-air pool on 2 February 2013 within the episode examined by prior meteorological and chemical studies [1,2]. All factorial comparisons use 14:00–18:00 UTC, corresponding to 07:00–11:00 mountain standard time. The first two forecast hours were excluded as spin-up, leaving five hourly samples; the sensitivity of the results to that two-hour choice has not been evaluated. Surface observations were extracted from the Synoptic archive for eleven stations reporting 2 m temperature, ten reporting 10 m wind speed, and nine reporting surface pressure in every retained hour. Subhourly reports were averaged within each UTC clock hour. The stations span basin-floor, sidewall, and bench locations but are neither spatially uniform nor area weighted. The 1 km second domain, d02, was selected because the 333 m third domain excludes

most higher bench stations. Figure 1 shows the domain geometry, the reporting stations, and the cross-basin transect used by the vertical-section diagnostics; the exclusion of most bench and basin-end stations from d03 is the concrete reason d02 is the common evaluation grid (ledger row C03). Four elevation-spanning stations provide surface context from 31 January 00:00 UTC through 4 February 00:00 UTC. Observed and modeled surface potential temperature for a two-site structural index was calculated as $\theta = T(1000/p)^{0.2854}$, with T in kelvin and p in hPa, and the index was defined as Mountain Home on the northwest bench minus Pariette Draw on the basin floor. Because the two sites are horizontally separated and locally exposed, the index is a spatial surface contrast rather than a colocated free-air inversion strength or depth. No sounding, profiler, or remote-sensing profile is used to validate the modeled 50–500 m layer.

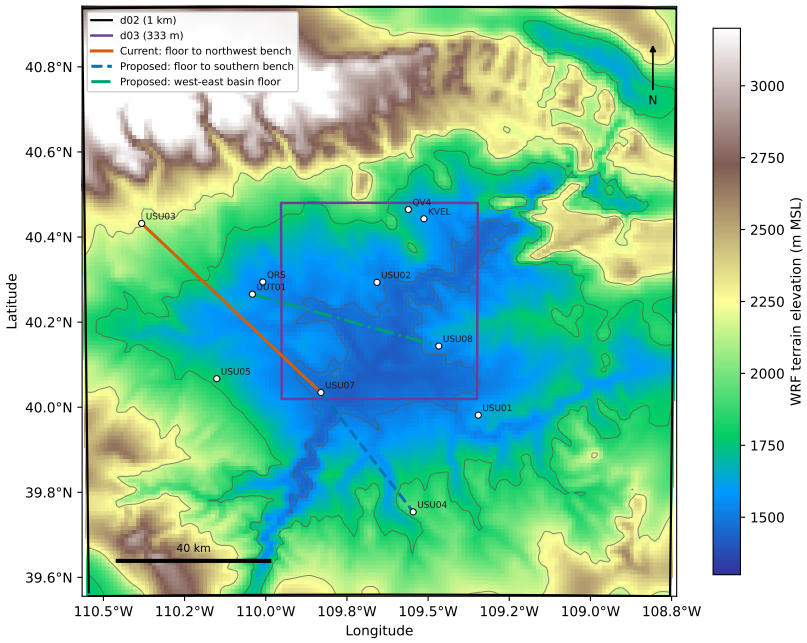


Figure 1. Terrain, reporting stations, and the cross-basin transect on the d02 evaluation domain, with the d03 interior domain outlined. Registered candidate figure (tracked with full provenance, not promoted). Most higher bench and basin-end stations fall outside d03, which is why d02 is the common evaluation domain (C03).

[Rewrite gate: Synoptic citation wording, the current API defaults, access conditions, and the frozen-cache hash are documented. Before circulation, archive or regenerate the historical request with explicit quality-control parameters, station and network ownership, sensor metadata, completeness screening, uncertainty treatment, request URL, and provider-specific reuse permission.]

2.3. WRF Configuration

The four free forecasts used WRF version 4.8.0 from 12:00 to 18:00 UTC on 2 February 2013. Three nested domains had horizontal grid spacing of 3, 1, and 0.333 km, 75 eta levels, and a 50 hPa model top. The common physics suite used Thompson microphysics, RRTMG longwave and shortwave radiation, the revised MM5 surface layer, the Noah land-surface model, and the YSU planetary-boundary-layer scheme. Grid and observation four-dimensional data assimilation were disabled. The runs shared domain geometry, default 30-arc-second-class terrain, physics, initialization time, output cadence, and valid window. The GFS one-way member reused the six verified metgrid files from the GFS two-

way member before rerunning `real.exe` and `wrf.exe`. The three-domain nest geometry common to the experiment family is shown in Figure 2.

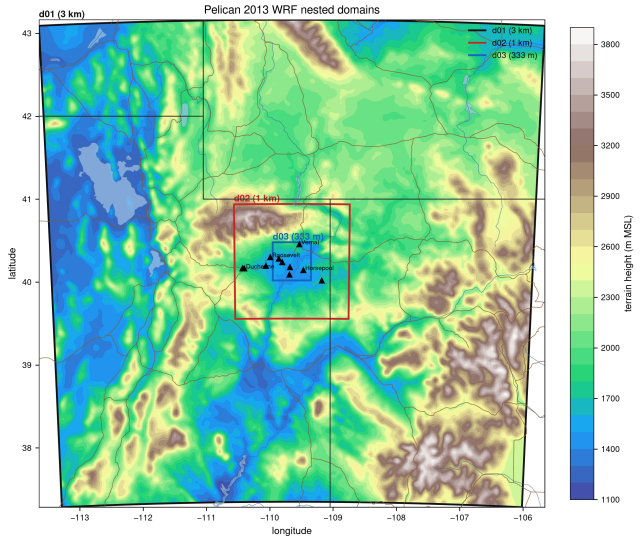


Figure 2. [Exploratory render.] Nested domain geometry for the experiment family: 3 km outer domain, 1 km second domain (d02, the common evaluation grid), and 333 m interior domain (d03). Batch engine render catalogued in `figures-draft/HARVEST.md`; configuration facts in the text come from the run inventory, not from this image.

[Rewrite gate: The WRF Version 4 technical reference and v4.8.0 release, GFS grid 004, and NAM grid 218 identities are verified. Insert only the keys generated by John's small curated bibliography export, and recover the exact input filenames and hashes, map projection, domain dimensions and centers, time step, boundary interval, static-geography versions, and snow-field initialization from the archived as-run records.]

2.4. The Archived Experiment Family and Its Controlled Comparisons

The four factorial members are the audited minimum of a larger archived family, X0–X10, built with the same WRF version, case, and initialization time (Table 2). Within that family, as-run namelist differencing establishes several additional controlled pairs: slope-aware radiation and topographic shading only (X1 versus X0), boundary-layer and surface-layer scheme only (X2 versus X1), static terrain source only (X3, X0, and X4 pairwise), and nest feedback at the fine-terrain control (X9 versus X0). Two further configurations, X7 and X8, differ from the rest in several dimensions at once and enter this study only as design context (Section 3.12). Table 3 lists the comparison identifiers used throughout, their authoritative differencing orientation, and their current evidence status. Run identity, exact archive paths, namelist fingerprints, and feedback flags for every member are recorded in the study's run inventory, and each comparison row resolves to those runs; directory or figure labels are never used as identity.

Table 2. The archived experiment family. Every run uses WRF 4.8.0, three nests at 3/1/0.333 km with 75 levels, Thompson microphysics, RRTMG radiation, and Noah land surface unless stated. "Fine terrain" is a 3-arc-second static source, "default" a 30-arc-second-class source, and "coarse" a 5-arc-minute source; source resolution is not model-grid spacing. X7 and X8 are context only: they differ from the controls in several dimensions at once.

Code	Product, feedback	Deliberate configuration role	Status
X0	NAM, one-way	fine-terrain control, YSU/MM5	control for X1, X3, X4, X9
X1	NAM, one-way	X0 plus slope radiation and shading	controlled pair (CMP07)
X2	NAM, one-way	X1 with MYJ/Eta instead of YSU/MM5	controlled pair (CMP08)
X3	NAM, one-way	default terrain; factorial member	pairs CMP02, 03, 09, 10
X4	NAM, one-way	deliberately coarse 5-arc-minute terrain	provenance caveat (CMP10, 11)
X5	NAM, two-way	default terrain; factorial member	pairs CMP01, 03
X6	GFS, two-way	default terrain; factorial member	pairs CMP01, 04
X7	NAM, two-way	1 km inner grid, 24 h integration	context only (CMP12)
X8	NAM, one-way	four domains to 111 m, 50 levels	context only (CMP12)
X9	NAM, two-way	X0 with two-way feedback	controlled pair (CMP06)
X10	GFS, one-way	default terrain; factorial member	pairs CMP02, 04

Table 3. Controlled comparisons referenced in this draft, their authoritative differencing orientation, and current status. "Confirmed" means a ledgered claim supports scoped prose now; "analyzed" means the matched recomputation is complete and committed; "diagnostic" means figures exist but no gate-passed quantitative claim; "context" means the comparison is not controlled. The archive figure engine renders the two default-terrain feedback families in the reverse orientation (two-way minus one-way); affected captions state this.

ID	Operation	Isolated lever	Status and gate
CMP01	X6 − X5	driving product (two-way)	confirmed (C04, C17, C25); spatial gate executed and failed (C30)
CMP02	X10 − X3	driving product (one-way)	confirmed (C04, C17, C25); spatial gate executed and failed (C30)
CMP03	X3 − X5	feedback, NAM, default terrain	confirmed near-null (C18, C25)
CMP04	X10 − X6	feedback, GFS, default terrain	confirmed small (C18, C25)
CMP05	(X10−X6)−(X3−X5)	feedback-by-product interaction	confirmed small (C18, C25)
CMP06	X9 − X0	feedback, NAM, fine terrain	gate passed; confirmed bound (C28; Section 3.8)
CMP07	X1 − X0	slope radiation and shading	screen executed and failed on a measurement-limited criterion (C31)
CMP08	X2 − X1	PBL and surface-layer scheme	confirmed station contrast (C32, refining C19); sole C6 candidate
CMP09	X3 − X0	terrain source, default vs fine	gate executed and failed (ANA-17; gate review)
CMP10	X3 − X4	terrain source, default vs coarse	controlled after lineage closure (C29); diagnostics committed, no gate declared
CMP11	X0 − X4	terrain source, fine vs coarse	controlled after lineage closure (C29); diagnostics committed, no gate declared
CMP12	X7 / X0 / X8	none (multiply confounded)	confirmed negative design finding (C16)

2.5. Surface and Common-AGL Diagnostics

Modeled 2 m temperature, earth-relative 10 m wind, and surface pressure were sampled at the nearest d02 cell to each station. Across the eleven fixed columns, model terrain minus station elevation had a median of -2.0 m and ranged from -32.4 to +14.3 m. That agreement limits one obvious mismatch but does not remove exposure, slope, land-cover, or network-coverage error. Mass-level height above ground was calculated from perturbation and base-state geopotential and the local WRF terrain height. Grid-relative winds were destaggered and rotated to earth-relative components. Potential temperature and both wind components were interpolated linearly at 25 m increments from 50 to 500 m above ground level (AGL) at each station column. Fixed-layer stability was defined as $S_{50:500} = \theta(500\,\mathrm{m}) - \theta(50\,\mathrm{m})$. The transport-wind proxy was $U_{50:500} = \{\bar{u}^2 + \bar{v}^2\}^{1/2}$, where the bars denote height means from 50 to 500 m AGL. This wind metric is not flow through the basin boundary, pollutant residence time, cold-pool volume, or a chemistry-model ventilation coefficient. Native WRF planetary-boundary-layer (PBL) height was retained only as a scheme-dependent companion diagnostic.

2.6. Footprint-Integrated Heat Deficit

The original basin-volume heat deficit of Whiteman et al. represents the sensible heat required to remove a basin inversion, and Adler et al. provide the closest published column implementation to the present diagnostic. **[Rewrite gate: Insert the Paperpile-generated keys for Whiteman et al. (1999) and Adler et al. (2023) after John's small curated export; do not invent or hand-edit keys.]** The present area-normalized atmospheric heat-deficit proxy was defined as $H = (c_p/g) \int \max\{\theta_{\mathrm{crest}} - \theta, 0\}\, dp$, integrated from the surface to a fixed crest reference height, where θ_{crest} is the potential temperature interpolated to that height in each modeled column. H is reported as a cell-area-weighted mean over an evaluation footprint in MJ m^{-2}. The footprint is the convex hull of the eleven evaluation stations and eleven registered basin waypoints, intersected with model terrain below the crest reference. The primary definition uses a 2200 m crest and the combined hull, covering 4965 d02 cells and 5277 km^2. Six alternate definitions vary the crest to 2100 and 2300 m, dilate or erode the boundary by one cell, or restrict the hull to the stations or waypoints alone. Because two-way feedback modifies the d02 terrain field by up to 23.1 m relative to the static one-way field, differences were formed within feedback mode, X6 minus X5 and X10 minus X3, on a common mask fixed from the one-way NAM terrain; the members of each pair share identical grids. Hourly values at 13:00 UTC provide trajectory context only, and all decisions use 14:00–18:00 UTC. Two criteria were fixed before inspection: a weaker-state criterion requiring both GFS-minus-NAM pairs to be negative at every retained hour under all seven definitions, and a faster-loss criterion requiring both GFS 14:00-to-18:00 UTC changes to be negative and more negative than the matched NAM changes under all seven definitions. The fixed 2200 m crest, nonnegative clipping, convex-hull footprint, and modeled profiles depart from the published definitions. The footprint is an evaluation region rather than a hydrologic basin boundary, and the proxy is neither a direct reproduction of the published diagnostics nor an observed cold-pool measurement.

2.7. Diagnostics for the Expanded Comparisons and Their Status

Three further diagnostic classes appear in this working revision. First, the fine-terrain feedback pair (CMP06) was recomputed with exactly the surface and common-AGL pipeline of the factorial: the same eleven station columns, hours, 50–500 m interpolation, pairing order, spatial-median reduction, and two-hour circular block resample, applied to the X9 and X0 archive outputs. The recomputation is committed with its exact command, inputs, and output fingerprints (job 13948333; `verification/data/cmp06_*.csv`), and its

orientation is the authoritative X9 minus X0, that is, two-way minus one-way, the reverse of the convention used for the default-terrain feedback rows. Second, a factorial effect-size panel (Figure 6) redraws the committed pair and interaction summaries; it introduces no new computation. Third, plan-view 2 m potential-temperature difference maps, east–west and north–south potential-temperature difference sections, and heat-deficit difference maps exist for every controlled pair as batch renders from the study's figure engine. These renders carry batch-level configuration provenance rather than per-image sidecars, are tracked in `figures-draft/HARVEST.md` for this draft, and have not passed the per-figure registration, inspection, and promotion gates that the promoted figures satisfy. They are therefore reproduced here as visual leads only, marked *exploratory render*, and no number in this draft is read from them. Fourth, the gate analyses themselves were executed for this revision with committed scripts whose criteria were fixed before any output was read: a spatial/vertical coherence analysis over the registered ANA-16 footprint with fixed display hours (ANA-19; job 13948410), a terrain-geometry analysis with a two-part X4 lineage check, predeclared elevation bands, and terrain-correlation criteria (ANA-17; job 13948411), and a two-leg process screen combining aspect-stratified radiation diagnostics (ANA-18; job 13948412) with the frozen station pipeline applied to both process pairs (job 13948378). Their summary tables, criteria verdicts, exact input hashes, and registered diagnostic figures are committed with the draft, and each Results subsection below reports its gate outcome explicitly.

2.8. Pairing, Aggregation, and Short-Sample Scope

Run differences were paired by station and valid hour before spatial aggregation. The forcing-by-feedback interaction was defined as $[\mathrm{GFS}_{1\mathrm{w}} - \mathrm{GFS}_{2\mathrm{w}}] - [\mathrm{NAM}_{1\mathrm{w}} - \mathrm{NAM}_{2\mathrm{w}}]$ and was likewise evaluated at each station and hour before spatial aggregation. Each station field was reduced to one unweighted spatial median per hour, and the five hourly medians were averaged for the reported effect. The stations were not treated as independent replicates. Hourly ranges and sign counts describe temporal consistency directly. A two-hour circular moving-block resample of the five hourly medians, repeated 20,000 times, and leave-one-hour-out ranges were retained in the reproducibility record as descriptive sensitivity checks. They are not confidence intervals, hypothesis tests, or estimates of observation, forcing, or structural uncertainty. The heat-deficit criteria in Section 2.6 were likewise fixed before any model output was inspected. No population inference is attempted from four deterministic integrations of one case.

3. Results

3.1. Observed Surface Context

The basin-floor station at Pariette Draw repeatedly cooled below the higher sites from 31 January through 3 February, while the magnitude and ordering of the station temperatures changed through the diurnal cycle (Figure 3). The recurring floor-to-bench separation is consistent with cold-pool conditions around the modeled period. It does not determine the cold-pool top, depth, volume, or vertical temperature structure.

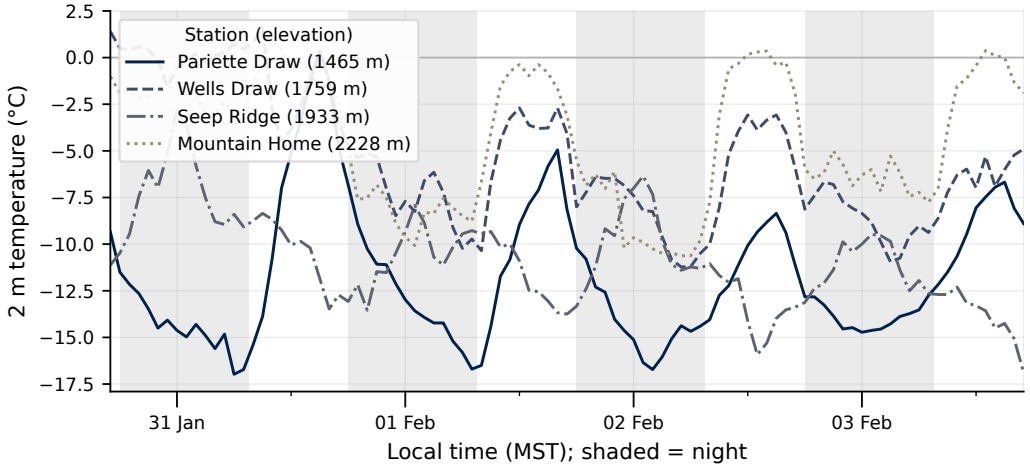


Figure 3. Observed 2 m-temperature evolution at four elevation-spanning Uinta Basin stations from 31 January through 3 February 2013. Shading indicates approximate local night. The sites establish a recurring spatial surface-temperature pattern but do not constitute a colocated vertical profile.

A station pseudo-profile view of the same observations, ordering the reporting sites by elevation, is retained as a registered candidate (Figure 4). It makes the floor-to-bench thermal structure legible at single hours, but the sites are horizontally separated by tens of kilometers, so the display is an apparent surface structure across terrain, not an observed free-air profile, inversion depth, or cold-pool top.

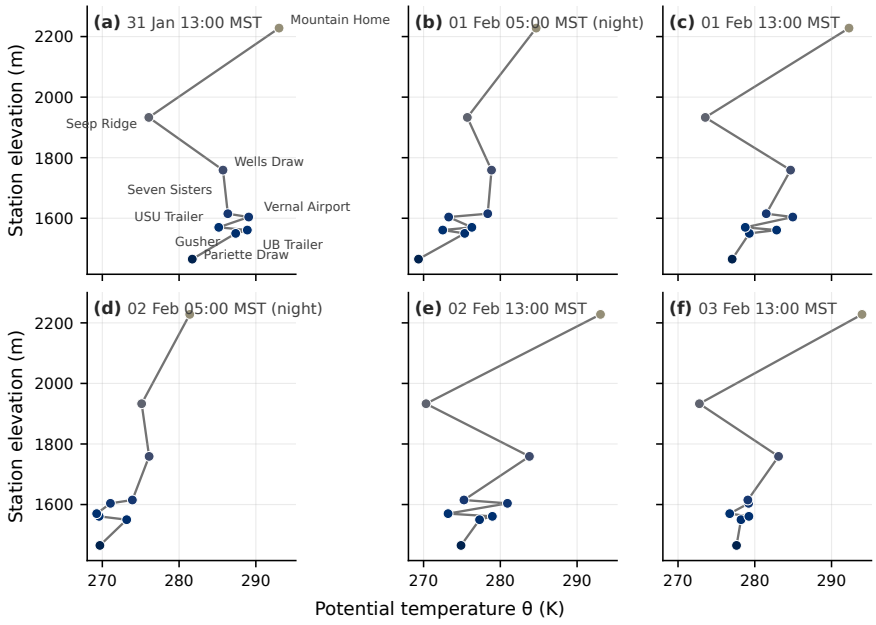


Figure 4. Observed surface potential temperature at the reporting stations arranged by station elevation at selected hours (station pseudo-profiles). Registered candidate figure. The sites are horizontally separated surface stations; the display is an apparent structure across terrain and must not be read as a vertical sounding.

3.2. Driving-Product Response in the Common Lower Layer

The common-AGL diagnostics separated the driving products under both feedback settings (Figure 5). GFS minus NAM reduced 50–500 m potential-temperature stability by 3.63 K under two-way feedback and 3.65 K under one-way feedback (Table 4). The five hourly spatial-median differences ranged from -5.58 to -1.13 K and from -5.36 to -0.90 K, respectively. All ten product-by-hour stability contrasts therefore retained the same

sign. GFS minus NAM increased fixed-layer transport wind by 0.79 m s^{-1} under two-way feedback and 0.99 m s^{-1} under one-way feedback. The hourly differences ranged from +0.37 to +1.18 m s^{-1} and from +0.61 to +1.50 m s^{-1}, with the same sign in every hour. Native PBL-height differences changed sign within both feedback modes and did not reproduce the stability and wind separation.

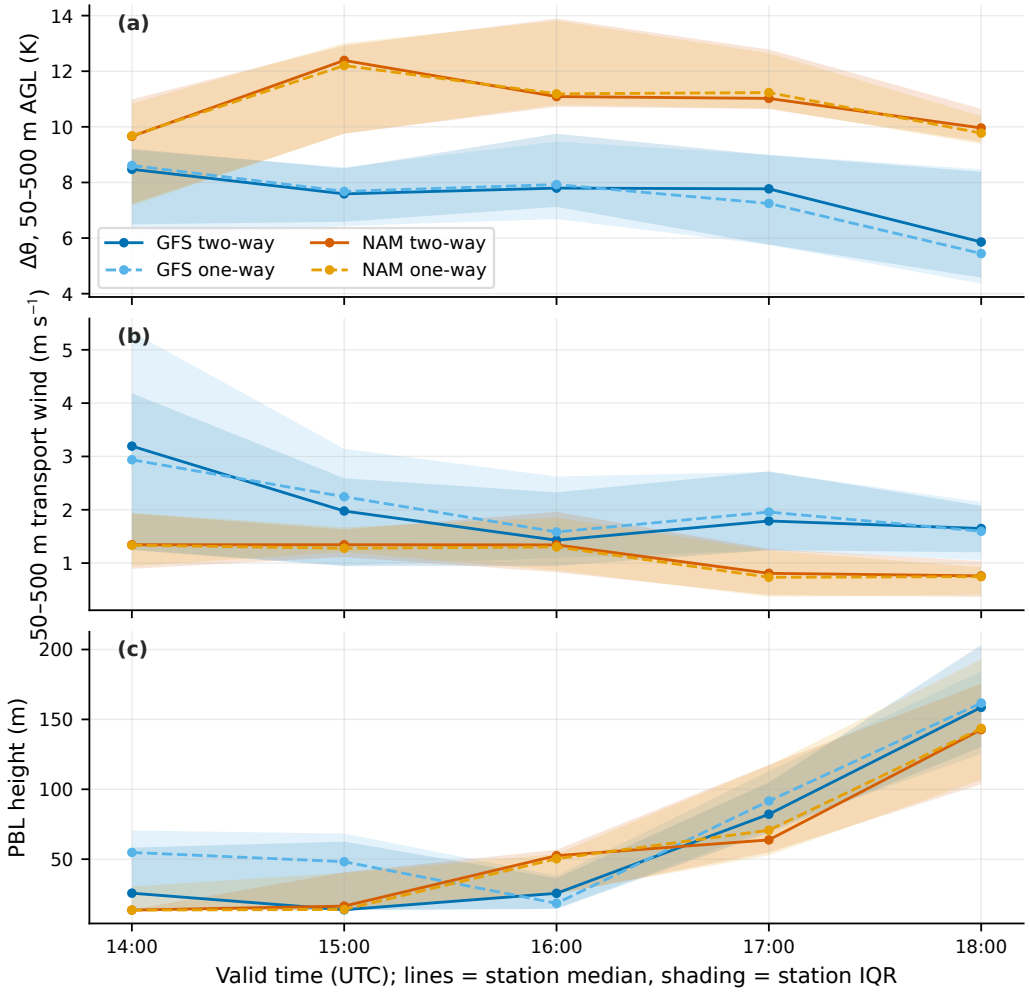


Figure 5. Station-column medians and interquartile ranges for 50–500 m AGL potential-temperature stability, fixed-layer transport wind, and native WRF PBL height over the matched window. Solid lines denote two-way feedback and dashed lines denote one-way feedback. The lower-layer GFS–NAM separation recurs under both feedback modes, while PBL height is retained as a scheme-dependent companion rather than an observed layer depth.

Table 4. Paired factorial effects averaged over the five hourly spatial medians. Positive temperature values indicate a warmer model-minus-observation bias, positive stability values indicate a more stable fixed layer, and positive transport-wind values indicate a larger layer-mean wind. The effects are deterministic descriptive contrasts rather than inferential estimates.

Effect	ΔT2 bias (°C)	$\Delta S_{50:500}$ (K)	$\Delta U_{50:500}$ (m s^{-1})
GFS − NAM, two-way	+1.79	-3.63	+0.79
GFS − NAM, one-way	+1.76	-3.65	+0.99
GFS one-way − two-way	+0.07	-0.07	+0.10
NAM one-way − two-way	-0.01	+0.00	+0.01
Feedback interaction	+0.04	-0.10	+0.10

3.3. The Factorial at a Glance

Figure 6 redraws the committed factorial summaries as one effect-size display: the two driving-product contrasts, the two default-terrain feedback contrasts, the fine-terrain feedback contrast of Section 3.8, and the paired interaction, for six metrics with their descriptive block-sensitivity intervals. The display adds no new computation; every dot and whisker is a committed summary value. Its single reading is scale separation: on every axis the driving-product rows sit far from zero relative to their sensitivity intervals, while all three feedback rows and the interaction are pinned near zero at the same scale. The surface-pressure panel also makes the metric disagreement of Section 3.4 visible: the pressure ordering favors GFS while the temperature ordering favors NAM.

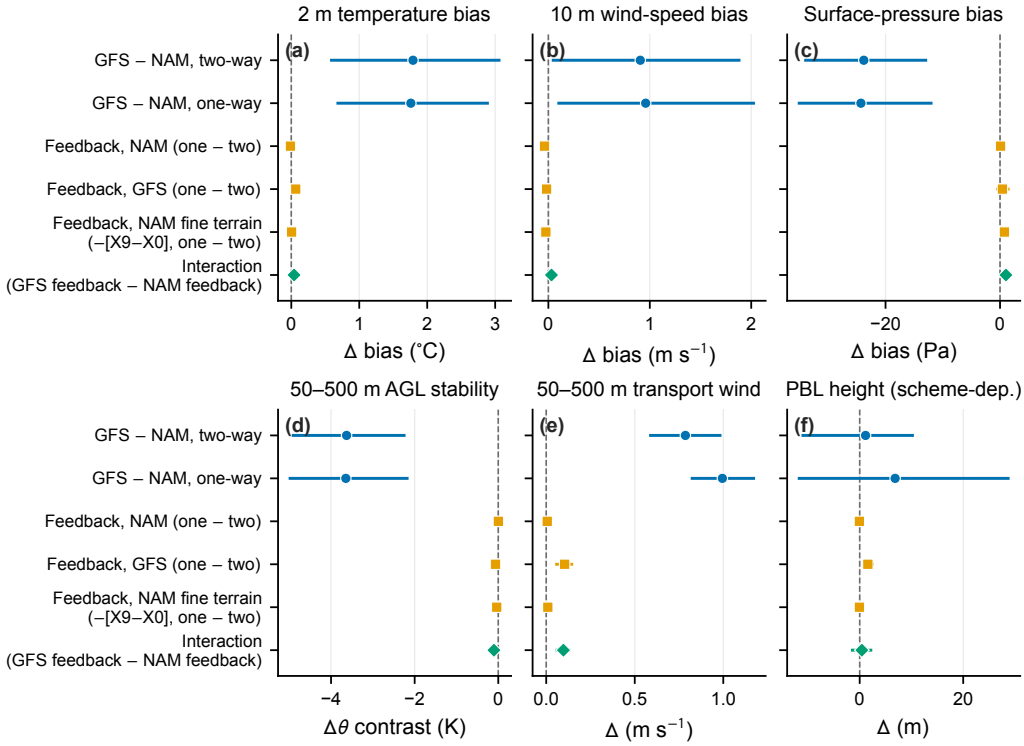


Figure 6. Factorial effect sizes from the committed pair and interaction summaries: mean five-hour effects (markers) and descriptive two-hour circular block-sensitivity intervals (whiskers) for six metrics. Blue circles are driving-product contrasts, orange squares feedback contrasts, and the green diamond the paired interaction; marker shape duplicates the color coding. The fine-terrain feedback row is the CMP06 recomputation (Section 3.8) negated into the one-way-minus-two-way orientation of the other feedback rows, as its label states. The intervals are sensitivity descriptions of five dependent hourly medians, not confidence intervals. Working figure drawn from committed data (`verification/scripts/plot_factorial_effects.py`); not yet registered or promoted.

3.4. Surface Corroboration and Metric-Dependent Skill

The paired surface-temperature result had the same product ordering as the layer diagnostics. Mean hourly station-median 2 m-temperature biases were +2.09 and +2.51 °C for the GFS members, against +0.40 and +0.60 °C for the matched NAM members. After pairing by station and hour, GFS minus NAM increased the temperature-bias contrast by +1.79 °C under two-way feedback and +1.76 °C under one-way feedback, and the corresponding absolute-error differences were +1.41 and +1.50 °C (Table 4). GFS minus NAM also increased 10 m-wind bias by +0.91 and +0.96 m s^{-1}, although the hourly difference in absolute wind error changed sign. Surface pressure ordered the products in the opposite direction. GFS minus NAM changed the paired surface-pressure absolute

error by -13.6 Pa under two-way feedback and -10.9 Pa under one-way feedback, negative in all five hours, while all four members retained a positive mean pressure bias between +43 and +56 Pa on the nine-station sample. These pressure differences are small in absolute terms, but their consistent sign shows that the surface metrics do not agree on a product ranking. Aggregate surface skill is therefore metric-dependent for these simulations, and it cannot by itself establish which member supplies the more credible lower layer.

The two-site potential-temperature index provides the only direct observation-model comparison of lower-layer structure in this study. Under two-way feedback, the GFS-driven member had a smaller Mountain Home-minus-Pariette Draw contrast than the matched NAM-driven member in every retained hour. The observed spatial contrast ranged from 12.63 to 20.21 K, the NAM member from 16.44 to 18.07 K, and the GFS member from 7.35 to 11.74 K. These two-site values combine elevation, horizontal distance, site exposure, and local surface energy balance, and they are not interpreted as free-air inversion depths.

[Rewrite gate: The existing bulk-theta render is withheld because its 8 K line came from a non-equivalent crest-relative sounding diagnostic. On CHPC, regenerate the figure from the frozen real-observation cache with the updated line-free script, update hashes in both manifests, inspect the output, and re-promote it before restoring the figure here.]

3.5. Where the Driving-Product Response Sits: Maps, Sections, and Upper-Air Context

[Diagnostic — not gate-passed: Gate executed with a negative outcome. The ANA-19 spatial-coherence analysis ran with predeclared criteria (job 13948410; `verification/data/spati` ledger row C30): the shallow-confinement and fixed-column-consistency criteria passed at all ten pair-hours, but the floor sign-coherence criterion failed at 14:00–15:00 UTC, so the predeclared conjunction fails and no spatial figure is promoted. The maps and sections below are registered diagnostic candidates and the engine renders remain exploratory; all numbers come from the committed summary tables, none from the images.]

The executed analysis quantifies the visual lead. Over the registered evaluation footprint, the cell-area-weighted mean GFS-minus-NAM 2 m potential-temperature difference is positive at every retained hour under both feedback modes, growing from +0.89–+1.16 K at 14:00–15:00 UTC to +2.00–+2.74 K at 16:00–18:00 UTC (Figure 7). The footprint fraction sharing the median sign is 0.64–0.67 in the first two retained hours — below the predeclared two-thirds threshold, which is why the gate fails — and 0.84–0.93 at 16:00–18:00 UTC. Vertical confinement passes at every pair-hour: the footprint mean of the absolute potential-temperature difference averaged over 50–500 m AGL exceeds the same quantity for 1500–2000 m AGL by factors of 1.45 to 3.12, and the ratio grows through the window (Figure 8). The footprint-mean 50–500 m stability difference is negative at every pair-hour, from -3.04 to -4.33 K, matching the fixed-column result (C17), and the 1500–2000 m layer-mean difference itself turns weakly negative (-0.16 to -0.42 K) in the last two hours while the shallow layer stays strongly positive. The registered transect sections (Figure 10) show the strongest warming in the lowest few hundred meters above the floor, a cool band near rim level, and a secondary elevated warm layer near 2500–3200 m MSL that remains an unanalyzed diagnostic lead. Because the predeclared floor-coherence criterion failed at the first two retained hours, Main C3 stays empty for this window; the committed summaries nevertheless document mean-sign coherence and shallow confinement explicitly, and the failure margin (0.001–0.025 below threshold) is recorded rather than argued away.

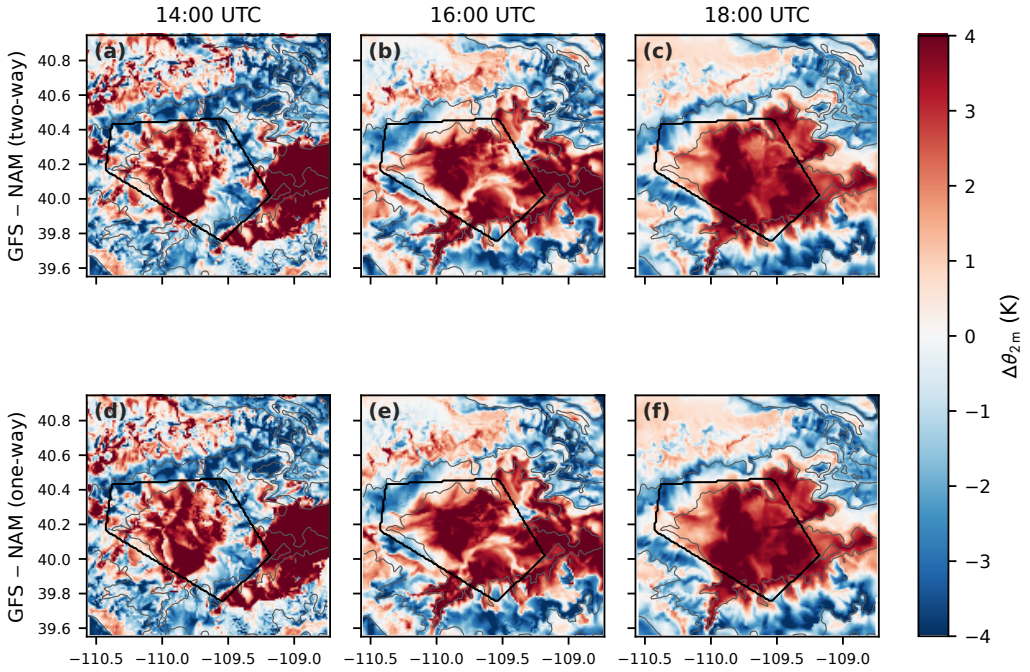


Figure 7. GFS-minus-NAM 2 m potential-temperature difference at the fixed display hours (14:00, 16:00, 18:00 UTC) under two-way (top, X6−X5) and one-way (bottom, X10−X3) feedback, with the registered evaluation footprint outlined in black and the 1700/2200 m terrain contours in gray. Registered diagnostic candidate from the executed ANA-19 analysis (job 13948410); not promoted because the predeclared floor-coherence criterion failed at 14:00–15:00 UTC (C30).

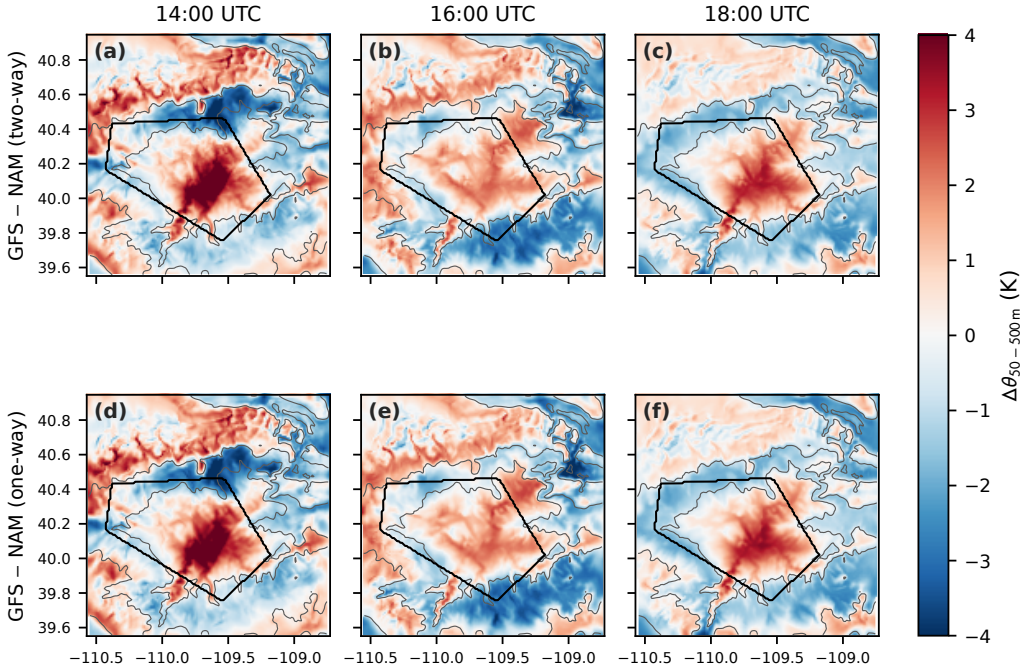


Figure 8. As Figure 7, but for the 50–500 m AGL layer-mean potential-temperature difference, the layer of the promoted fixed-column diagnostics. Registered diagnostic candidate (ANA-19, job 13948410); the shallow-confinement criterion computed from this layer passed at all ten pair-hours.

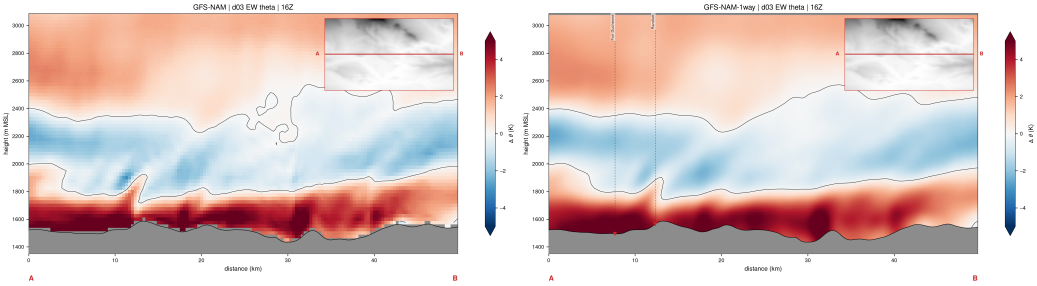


Figure 9. [Exploratory render.] East–west potential-temperature difference sections across the basin at 16:00 UTC for the two-way (left) and one-way (right) GFS-minus-NAM pairs. Batch engine renders retained for comparison with the registered transect sections of Figure 10.

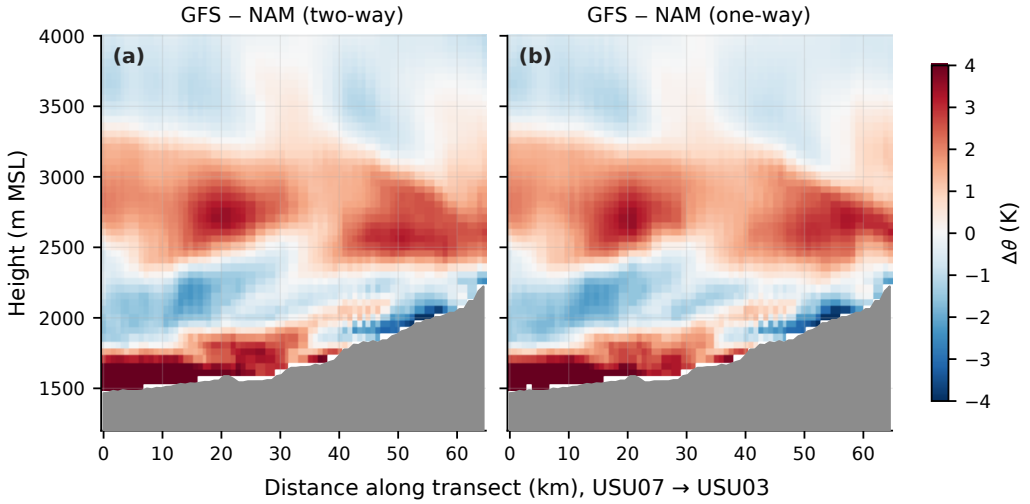


Figure 10. Registered diagnostic transect sections (ANA-19, job 13948410): GFS-minus-NAM potential-temperature difference along the USU07-to-USU03 transect at 16:00 UTC, the predeclared middle hour of the decision window, for the two-way (left) and one-way (right) pairs. The strongest warming sits in the lowest few hundred meters, with a cool band near rim level and a secondary elevated warm layer near 2500–3200 m MSL that is an unanalyzed lead.

Above the basin, registered candidate difference maps at 700 and 500 hPa (Figure 11) carry the diagnostic lead that the GFS-minus-NAM difference is more coherent at 700 hPa at 14:00 and 16:00 UTC than the mixed, evolving 500 hPa pattern. Terrain intersects the 700 hPa surface around the basin rim, so the masked region prevents any simple causal reading; the panels are context for the shallow-response hypothesis, not evidence of it.

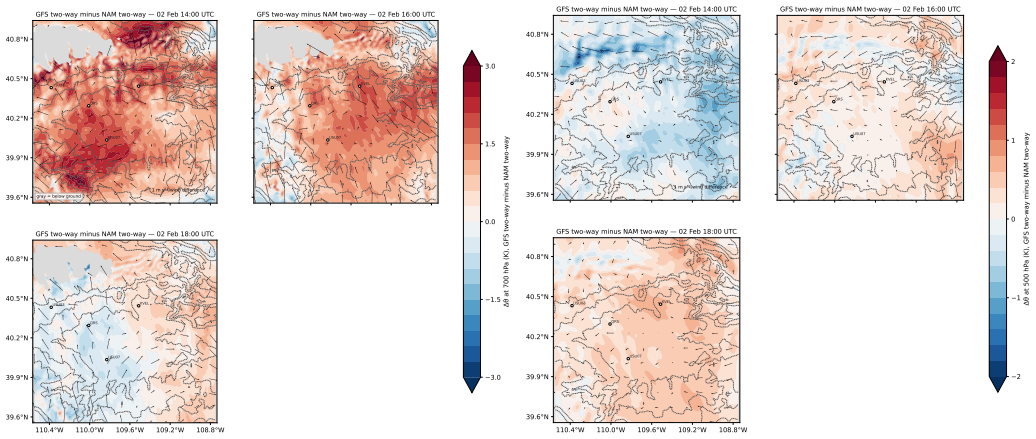


Figure 11. GFS-minus-NAM potential-temperature and wind differences at 700 hPa (left) and 500 hPa (right) for the two-way pair at selected matched hours. Registered candidate figures with exact run provenance; not promoted. Terrain masking near the rim limits interpretation at 700 hPa, and the panels serve as vertical context only.

3.6. Footprint Heat Deficit: A Weaker State without Faster Loss

Under the primary definition, the footprint-mean heat deficit of the GFS members rose from 2.68 to 3.00 MJ m^{-2} (two-way) and from 2.66 to 2.95 MJ m^{-2} (one-way) between 14:00 and 18:00 UTC, while the matched NAM members rose from 4.18 to 4.33 and from 4.19 to 4.32 MJ m^{-2} (Table 5 and Figure 12). The paired GFS-minus-NAM differences averaged -1.23 MJ m^{-2} under two-way feedback and -1.25 MJ m^{-2} under one-way feedback. All 70 retained pair-hour-definition differences were negative, spanning -1.80 to -0.70 MJ m^{-2}, so the weaker-state criterion passed in all 14 pair-definition combinations. The faster-loss criterion required an actual GFS decrease that was also larger than the matched NAM change. Instead, heat deficit increased in every member, by +0.32 and +0.30 MJ m^{-2} for GFS and by +0.15 and +0.13 MJ m^{-2} for NAM under two-way and one-way feedback. The difference in change was +0.17 MJ m^{-2} under both feedback modes for the primary definition and spanned -0.07 to +0.32 MJ m^{-2} across definitions, without a consistent sign. The faster-loss criterion therefore failed in all 14 pair-definition combinations. The GFS-minus-NAM separation consequently narrowed slightly over the window, from -1.50 to -1.33 MJ m^{-2} under two-way feedback and from -1.53 to -1.36 MJ m^{-2} under one-way feedback. The footprint result shows a consistently weaker simulated pool in the GFS members at every retained hour, not a divergence in its evolution.

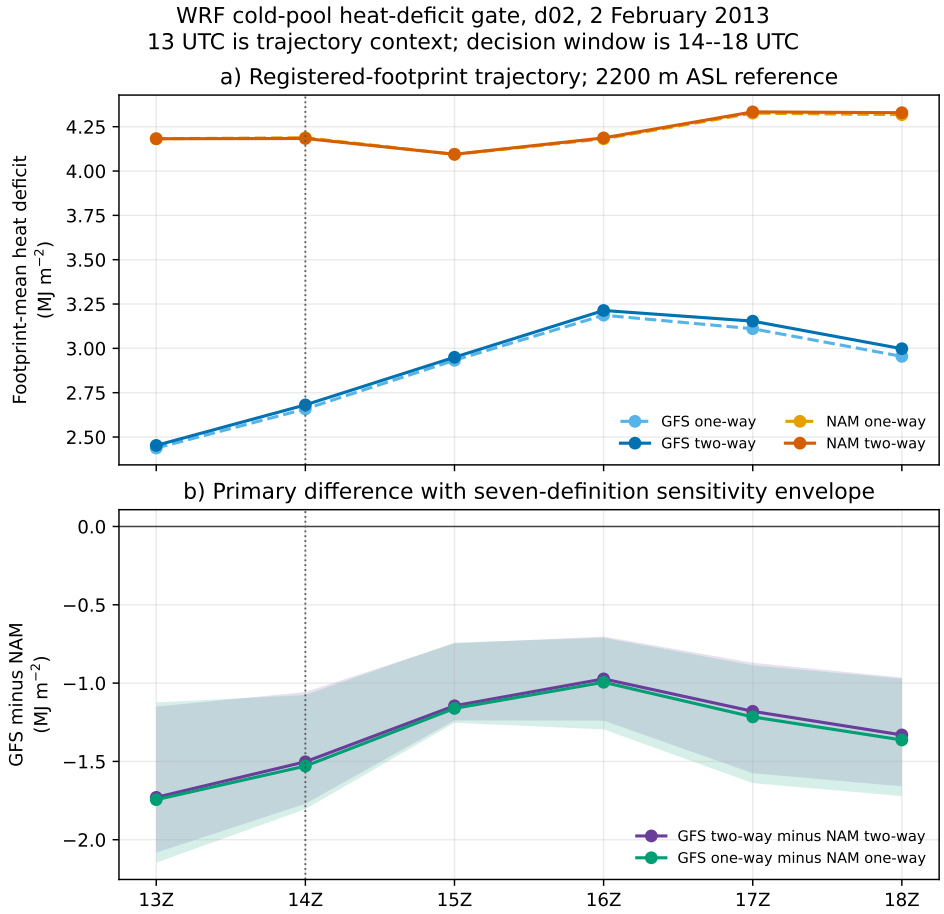


Figure 12. Footprint-mean heat-deficit trajectories for the four factorial members from 13:00 to 18:00 UTC under the primary footprint definition, with the seven-definition sensitivity envelopes for the two GFS-minus-NAM pairs. Registered candidate figure, visually inspected; the plotted values are the committed trajectory and sensitivity tables that also support Table 5. The 13:00 UTC values are trajectory context only; all criteria use 14:00–18:00 UTC.

Table 5. Footprint-integrated heat-deficit contrasts over the 14:00–18:00 UTC decision window. The primary definition uses the combined station-and-waypoint hull with a 2200 m crest (4965 cells, 5277 km^2); envelopes span all seven footprint definitions. The values are deterministic descriptive contrasts in MJ m^{-2}.

GFS − NAM quantity	Two-way pair	One-way pair
Mean difference (primary)	-1.23	-1.25
Hourly range (primary)	-1.50 to -0.97	-1.53 to -0.99
Hourly envelope (seven definitions)	-1.77 to -0.70	-1.80 to -0.71
GFS change, 14:00–18:00 UTC	+0.32	+0.30
NAM change, 14:00–18:00 UTC	+0.15	+0.13
Difference in change (primary)	+0.17	+0.17
Weaker-state criterion passed	7/7	7/7
Faster-loss criterion passed	0/7	0/7

The spatial counterpart of the footprint integral is retained as an exploratory lead (Figure 13): at 16:00 UTC the GFS-minus-NAM heat-deficit difference field is negative across the basin floor rather than concentrated at a few cells, visually consistent with the unanimous footprint result. An engine time-series view across the wider run family (Figure 14) likewise suggests that the driving-product separation exceeds the spread among

the NAM configuration variants, but its reduction differs from the audited footprint pipeline and it carries no claim.

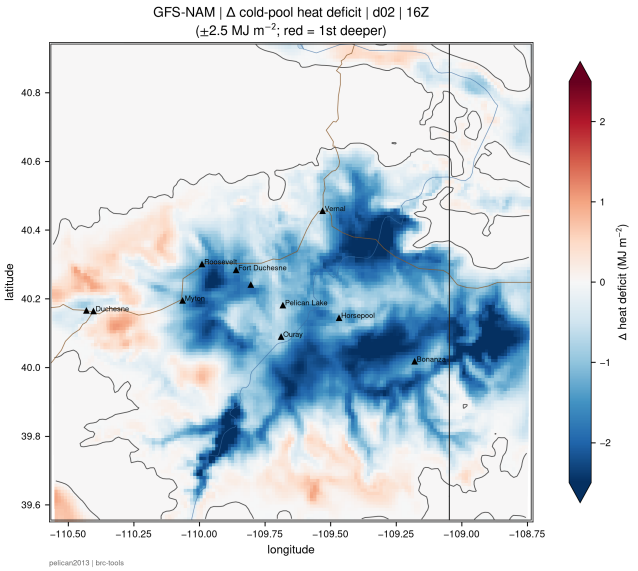


Figure 13. [Exploratory render.] GFS-minus-NAM (X6−X5) heat-deficit difference field at 16:00 UTC on a fixed ±2.5 MJ m^{-2} scale. Batch engine render; the audited quantitative result is the footprint integral of Table 5, not this image.

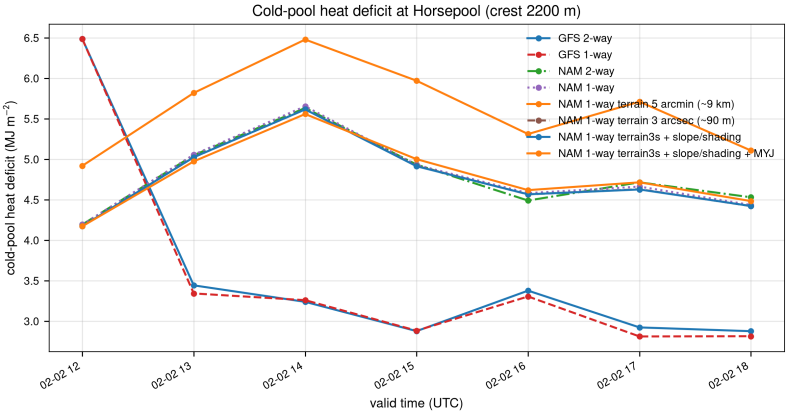


Figure 14. [Exploratory render.] Engine heat-deficit time series across the archived run family. The reduction behind this render differs from the audited footprint pipeline (different masks and averaging), so it is reproduced only as an archive-breadth illustration; no value in the text is read from it.

3.7. The Nest-Feedback Bound at Default Terrain

The within-product feedback responses were much smaller than the GFS-minus-NAM contrasts, but they were not identical between products (Table 4). For NAM, one-way minus two-way averaged +0.002 K for stability and +0.006 m s^{-1} for transport wind, and the hourly effects changed sign within -0.042 to +0.088 K and -0.035 to +0.033 m s^{-1}. For GFS, one-way minus two-way averaged -0.066 K for stability and +0.104 m s^{-1} for transport wind. The GFS stability effect ranged from -0.198 to +0.067 K and was negative in four of five hours, while the transport-wind effect ranged from +0.025 to +0.197 m s^{-1} and was positive in all five hours. The paired feedback interaction averaged -0.103 K for stability and +0.097 m s^{-1} for transport wind, with the wind component positive in all five hours. The 2 m-temperature-bias interaction was +0.041 °C, and the PBL-height interaction changed

sign from hour to hour. These values bound the response of the present configuration and sample; they do not establish general equivalence between feedback modes.

The corresponding plan-view differences are retained as exploratory context (Figure 15). Their visual lead matches the paired numbers: both default-terrain feedback fields are near null over the basin interior, with the stronger colors confined to terrain and domain-edge cells, and the GFS family shows slightly more coherent structure than the NAM family.

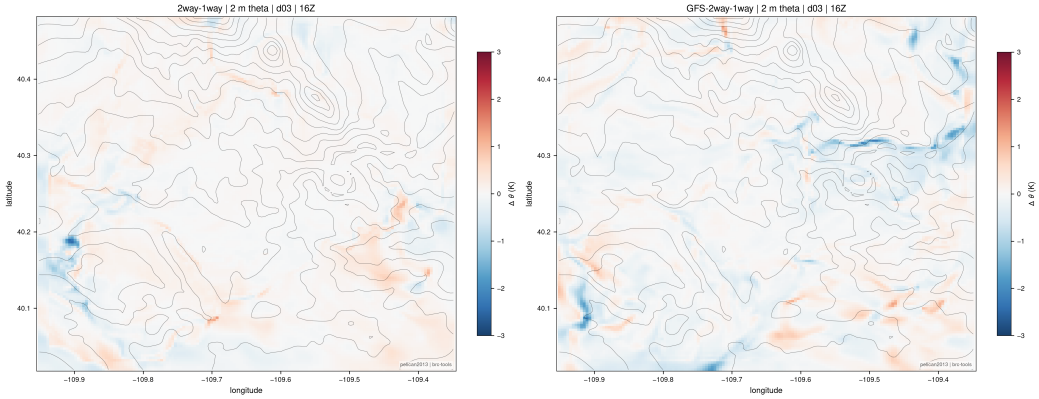


Figure 15. [Exploratory render.] Within-product feedback difference fields for 2 m potential temperature at 16:00 UTC: NAM family (left) and GFS family (right). The engine renders these fields as two-way minus one-way (X5−X3 and X6−X10), the *reverse* orientation of the paired rows CMP03 and CMP04 and of Table 4; signs must not be transferred between the two. Batch engine renders; visual context for the bounded feedback result.

3.8. The Feedback Bound Replicates at Fine Terrain (CMP06)

The fine-terrain control pair, X9 minus X0, isolates the same feedback flag as the default-terrain contrasts but at the 3-arc-second static terrain of the control configuration. Recomputed on exactly the factorial's matched surface and common-AGL definitions (Section 2.7), every fine-terrain feedback effect is again one to two orders of magnitude smaller than the matched driving-product contrast (Table 6). Two-way minus one-way stability averaged +0.04 K (hourly range -0.09 to +0.14 K, positive in four of five hours), transport wind -0.01 m s^{-1} (range -0.07 to +0.03 m s^{-1}, mixed signs), and 2 m-temperature bias -0.004 °C (range -0.09 to +0.04 °C, mixed signs). The block-sensitivity intervals for stability, transport wind, PBL height, and temperature bias all span zero. The two metrics with consistent hourly sign are small: 10 m-wind absolute error increased by +0.03 m s^{-1} in all five hours, and surface pressure decreased by -0.77 Pa in bias and absolute error in all five hours on the nine-station sample. For comparison at the same terrain and physics, the CMP01 and CMP02 driving-product contrasts move stability by more than 3.6 K and transport wind by 0.8 to 1.0 m s^{-1}. The bounded-feedback conclusion of Section 3.7 therefore survives the change of terrain regime. This recomputation passed its predeclared gate (the ANA-20 falsifier — a fine-terrain feedback effect approaching the driving-product magnitude — did not occur) and is recorded as ledger row C28; no figure is promoted, and the engine renders below remain exploratory.

Table 6. Fine-terrain feedback contrast CMP06 (X9 minus X0, *two-way minus one-way*, the authoritative orientation of the comparison inventory and the reverse of the feedback rows in Table 4). Values are means of five hourly spatial medians from the committed recomputation (job 13948333), with hourly ranges, sign counts, and descriptive two-hour block-sensitivity intervals. Sensitivity intervals are not confidence intervals.

Metric	Mean	Hourly range	Signs (+/−)	Sens. interval
Stability $S_{50:500}$ (K)	+0.04	-0.09 to +0.14	4/1	-0.03 to +0.10
Transport wind $U_{50:500}$ (m s^{-1})	-0.01	-0.07 to +0.03	3/2	-0.03 to +0.01
PBL height (m)	+0.05	-0.67 to +0.78	3/2	-0.38 to +0.47
T2 bias (°C)	-0.004	-0.09 to +0.04	3/2	-0.05 to +0.03
T2 absolute error (°C)	-0.006	-0.06 to +0.05	2/3	-0.04 to +0.03
10 m wind bias (m s^{-1})	+0.02	-0.01 to +0.06	4/1	+0.01 to +0.04
10 m wind absolute error (m s^{-1})	+0.03	+0.001 to +0.06	5/0	+0.01 to +0.06
Surface-pressure bias (Pa)	-0.77	-1.23 to -0.35	0/5	-1.06 to -0.50
Surface-pressure absolute error (Pa)	-0.77	-1.23 to -0.42	0/5	-1.01 to -0.59

The plan-view and heat-deficit difference fields for the fine-terrain pair carry the matching visual lead (Figure 16): on a deliberately tight ±0.2 MJ m^{-2} scale the heat-deficit difference appears as low-amplitude speckle over the basin floor rather than a coherent basin-scale signal, and the 2 m field is likewise near null away from terrain edges.

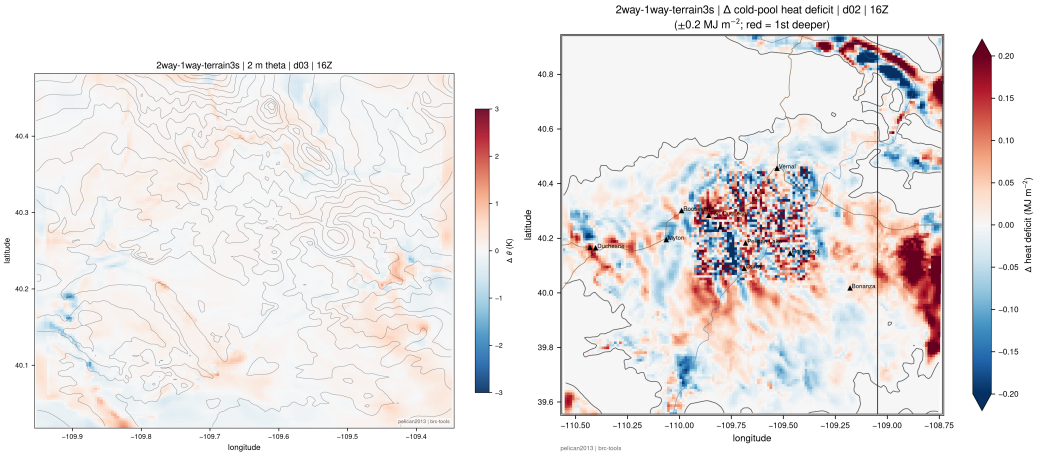


Figure 16. [Exploratory render.] Fine-terrain feedback difference fields (X9−X0, two-way minus one-way, matching the orientation of Table 6) at 16:00 UTC: 2 m potential temperature (left) and heat deficit on a fixed ±0.2 MJ m^{-2} scale (right). Batch engine renders; the quantitative result is Table 6.

3.9. Terrain-Source Brackets (CMP09–CMP11)

[Diagnostic — not gate-passed: Gate executed: CMP09 failed its predeclared criteria, and the X4 lineage check passed, closing that standing caveat (job 13948411; `verification/data/terrain_response_criteria.csv`; ledger row C29). No terrain slot is promoted; the quantitative values below are committed descriptive diagnostics, not gate-passed claims.]

The archive brackets the static terrain source on an otherwise identical NAM one-way configuration: default minus fine (CMP09, X3−X0), default minus coarse (CMP10, X3−X4), and fine minus coarse (CMP11, X0−X4). The executed analysis first resolves identity: the archived X4 `namelist.input` hashes byte-identically to the X0/X3 namelist, and its terrain is far smoother than the default member's, so the coarse-terrain member is confirmed as the same configuration family with only the static source changed (C29), and CMP10/CMP11 are controlled comparisons. The fine bracket itself is small on this grid: the X3-minus-X0 terrain difference has a root-mean-square of only 7.9 m (range -55 to +60 m), and its 2 m response is correspondingly small, with elevation-band means of 0.07–0.27 K (Figure 17). The response is slope-organized — the slopes-band mean exceeds the floor band by factors

of 1.7–2.4 at every hour, passing the first predeclared criterion — but its area-weighted correlation with the terrain difference stays below the predeclared 0.3 magnitude at 14:00–16:00 UTC (reaching 0.34–0.36 only late in the window), so the CMP09 gate fails and Main C5 stays empty for this window. The coarse brackets are an order of magnitude larger (terrain RMS 91.9 and 93.1 m, ranges to -462/+487 m) and their committed diagnostics are correspondingly stronger: slopes-band responses of 1.2–1.5 K, terrain correlations of 0.50–0.67 at every hour, and station-sample mean responses up to +0.61 K. No gate was predeclared for the coarse brackets, so those values carry no claim; declaring and testing one is now unblocked by the lineage closure. The 16:00 UTC plan views (Figure 18) show the same picture visually: the terrain-source response organizes along the basin rim, sidewall slopes, and tributary drainages rather than as a uniform basin-floor offset, and its apparent amplitude grows with the width of the source bracket. The east–west section for the fine bracket (Figure 19, left) suggests the same edge-concentrated structure in the vertical, and the unregistered local prototype (Figure 19, right) records the geometry-and-capacity hypothesis a future coarse-bracket gate would test. Per-station model-minus-station elevations for all three members are committed (`verification/data/terrain_station_elevation.csv`), making the pressure-artifact mechanism of Section 4.6 directly checkable.

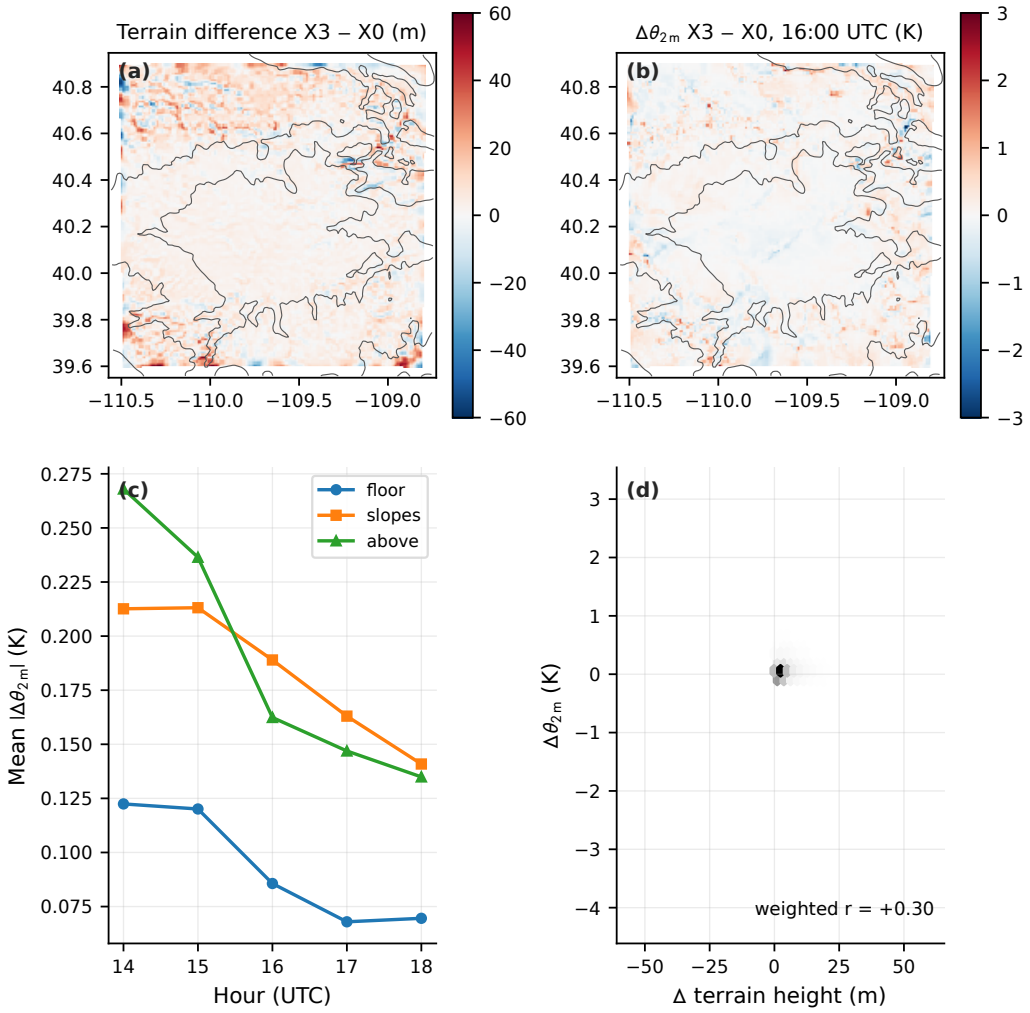


Figure 17. Registered diagnostic panel for the executed CMP09 analysis (ANA-17, job 13948411): (a) the X3−X0 terrain difference; (b) the 2 m potential-temperature response at the fixed 16:00 UTC display hour; (c) elevation-band means of the absolute response by hour; (d) the response-versus-terrain-difference density with the area-weighted correlation. The slope-organization criterion passed and the terrain-correlation criterion failed, so no promotion follows (gate review; C29 covers the lineage closure only).

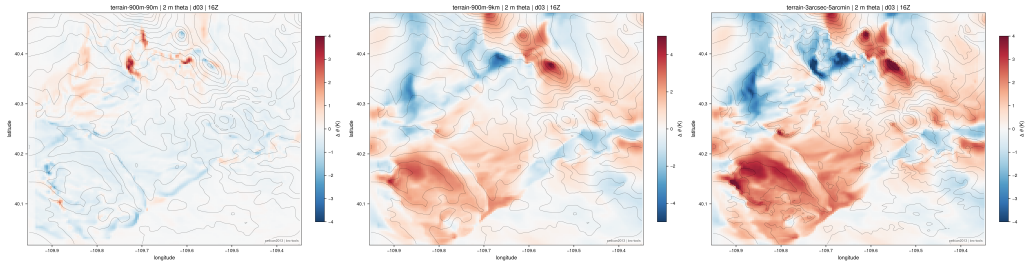


Figure 18. [Exploratory render.] Terrain-source difference fields for 2 m potential temperature at 16:00 UTC: default minus fine (left, CMP09, X3−X0), default minus coarse (center, CMP10, X3−X4), and fine minus coarse (right, CMP11, X0−X4). Batch engine renders; the X4 lineage is now resolved (C29), so all three are controlled comparisons, but only CMP09 has an executed (and failed) gate.

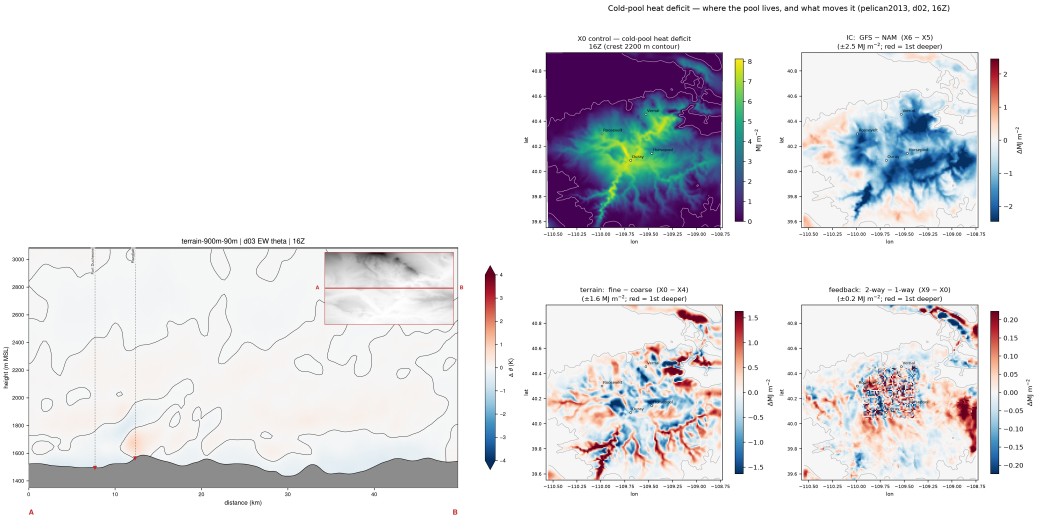


Figure 19. [Exploratory render.] Left: east–west potential-temperature difference section for the default-minus-fine terrain pair (CMP09) at 16:00 UTC, batch engine render. Right: unregistered local prototype overlaying the fine-minus-coarse heat-deficit difference with the registered evaluation footprint at 16:00 UTC; the prototype's thresholding is known to be sensitive and its interpretation is a hypothesis for ANA-17, not a result.

The heat-deficit counterpart for the widest bracket (Figure 20) shows the same rim-and-tributary organization on a fixed ± 1.6 MJ m^{-2} scale.

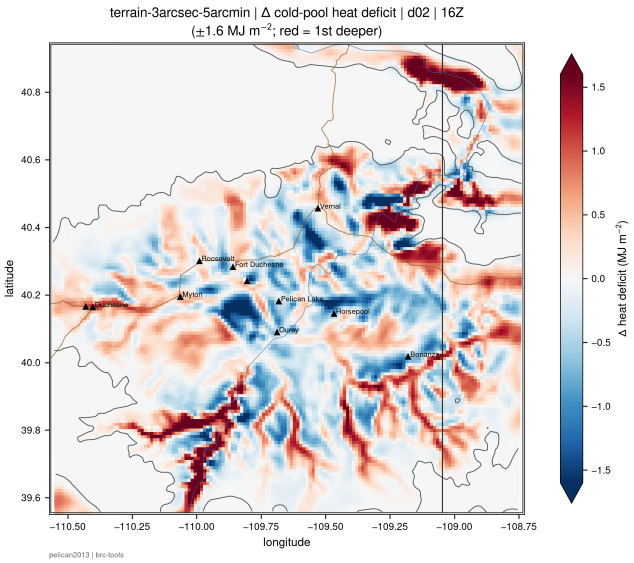


Figure 20. [Exploratory render.] Fine-minus-coarse (CMP11, X0−X4) heat-deficit difference at 16:00 UTC on a fixed ± 1.6 MJ m^{-2} scale. Batch engine render; controlled after the lineage closure (C29), with no declared gate yet.

3.10. Process Treatments: Slope-Aware Radiation and PBL/Surface-Layer Scheme

[Diagnostic — not gate-passed: Screen executed (jobs 13948412 and 13948378; ledger rows C31 and C32). CMP07 failed its predeclared shortwave criterion for a measurement-limited reason recorded below; CMP08 produced a coherent station-sample contrast on the frozen pipeline and is the sole remaining Main C6 candidate. The archived run summaries for X1 and X2 still expose no source-code commit, a standing provenance caveat.]

The slope-aware radiation pair (CMP07, X1−X0) changes only `slope_rad`, `topo_shading`, and `shadlen`, and the members are bitwise identical at initialization. The executed aspect-

stratified screen found that the archived downward-shortwave field does not respond at all — region means stay at or below 0.02 W m^{-2} and the south-minus-north facing contrast never exceeds 0.27 W m^{-2} in magnitude — because `slope_rad` modifies the solar flux inside the surface energy budget while the archived SWDOWN diagnostic remains the flat-plane value, and no surface heat fluxes were archived. The predeclared shortwave criterion therefore fails on a diagnostic the archive cannot express (C31), and no Main C6 promotion follows from CMP07. The surface nevertheless responds exactly where the treatment acts: skin temperature on south-facing steep cells warms relative to north-facing steep cells by +1.08 to +2.23 K at every post-sunrise hour, with the 2 m potential-temperature contrast at +0.16 to +0.90 K, an aspect-organized pattern that paints the sidewalls and the Uinta slopes while leaving the basin floor untouched (Figure 21). At the station columns the pair is essentially null (stability +0.004 K with a +0.001 to +0.007 block interval, transport wind +0.006 m s^{-1}, 2 m-temperature bias +0.001 °C; Table 7), confirming that the aggregate similarity reflects station placement on the floor rather than treatment inertness. The engine plan views near sunrise and late morning (Figure 22) and the low-amplitude, slope-organized heat-deficit difference (Figure 24, left) carry the matching visual leads.

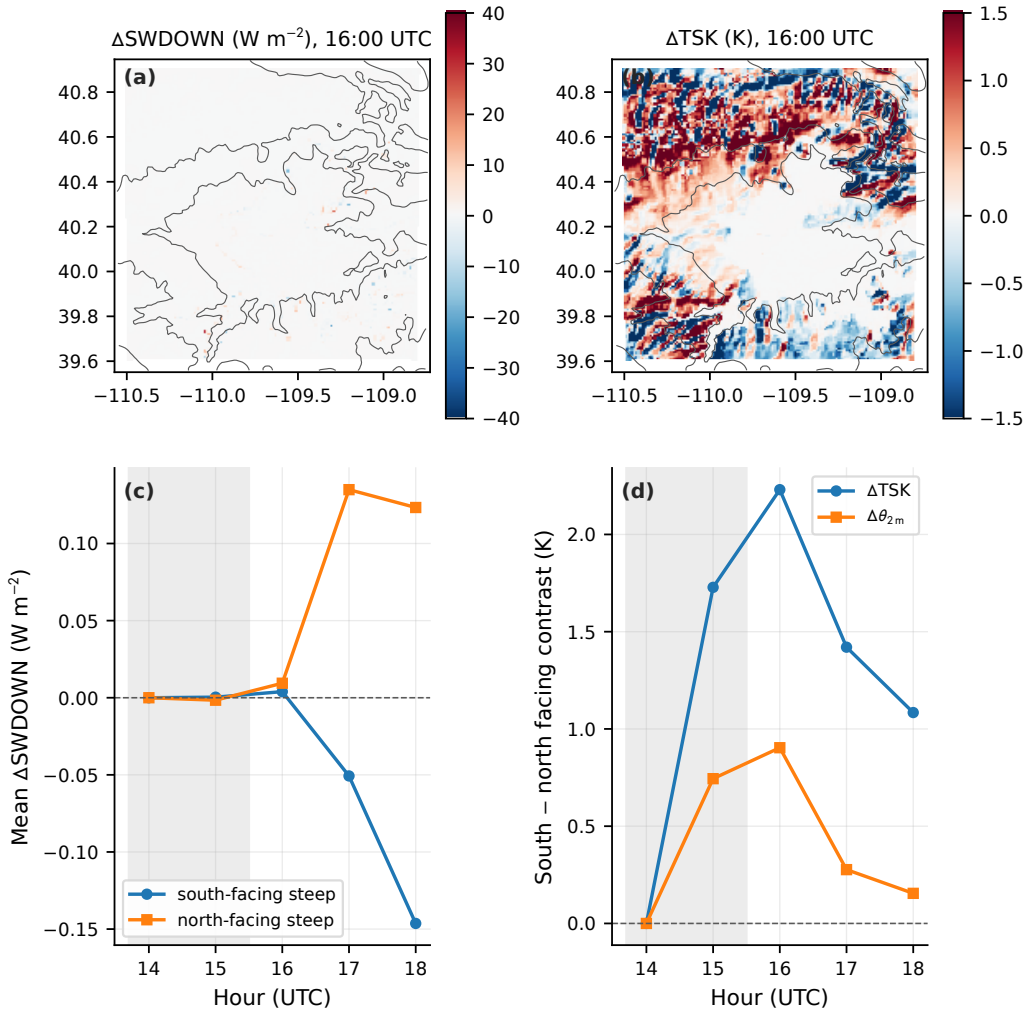


Figure 21. Registered diagnostic panel for the executed CMP07 screen (ANA-18, job 13948412): (a) the unresponsive flat-plane ΔSWDOWN field and (b) the aspect-organized ΔTSK field at 16:00 UTC; (c) mean ΔSWDOWN on south- and north-facing steep cells by hour; (d) the south-minus-north facing contrast in skin temperature and 2 m potential temperature. Shading marks the pre-/near-sunrise hours excluded from the decision criteria. The predeclared shortwave criterion failed for the recorded measurement-limited reason; the skin-temperature response is a committed diagnostic, not a gate-passed claim (C31).

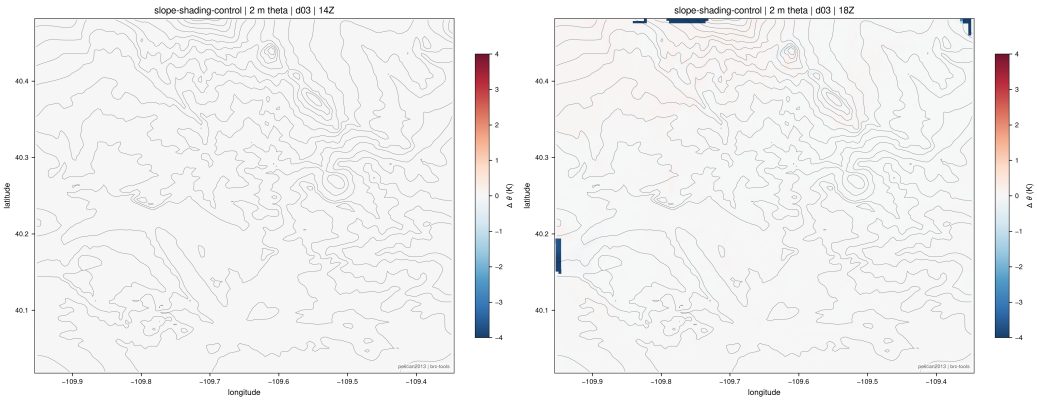


Figure 22. [Exploratory render.] Slope-radiation and shading response (CMP07, X1−X0) in 2 m potential temperature near sunrise at 14:00 UTC (left) and in late morning at 18:00 UTC (right). Batch engine renders behind gate ANA-18. The hours differ from the other map figures deliberately: the treatment is sun-angle dependent.

The scheme pair (CMP08, X2−X1) changes only the boundary-layer and surface-layer schemes, MYJ/Eta against YSU/MM5, while holding the slope-aware radiation configuration fixed. Recomputed on the frozen station pipeline, the pair produces the most coherent non-driving-product contrast in the archive (Table 7; C32): MYJ/Eta warms the 2 m-temperature bias by +0.37 °C in all five hours, reduces the 10 m-wind absolute error by 0.145 m s^{-1} in all five hours, raises the surface-pressure bias by +6.3 Pa in all five hours, and increases the fixed-layer transport wind by +0.098 m s^{-1} in all five hours with a block interval (+0.069 to +0.126) that excludes zero. The stability response is mixed (+0.11 K, three positive hours of five, interval spanning zero), and the PBL-height decrease of 9.2 m is scheme-defined and carries no structural meaning across schemes. These magnitudes sit one order below the driving-product contrast and one to two orders above the feedback bounds, and the sign split — temperature error worsens while wind error improves — is another in-experiment demonstration that aggregate surface skill is metric-dependent. This refines the ledger's provisional C19: the earlier two-site bulk contrast (18.49 K for X2 against 17.18 K for X1) is now backed by matched common-AGL and surface summaries, and the inherited description of a uniformly small sub-700 m scheme response is not supported. Its 16:00 UTC map and section (Figure 23) show the matching coherent low-level response, and the MYJ/Eta heat-deficit difference (Figure 24, right) is visibly stronger than the radiation counterpart, consistent with a process treatment that alters stable-layer mixing directly. On coherence and information gain the executed screen therefore selects CMP08 as the sole Main C6 candidate; whether it enters the main text or the supplement is an editorial decision at review, and it is not presented as a scheme skill ranking.

Table 7. Executed process-pair station contrasts on the frozen pipeline (14:00–18:00 UTC; job 13948378). Values are means of five hourly spatial medians; sign counts are positive/negative hours; intervals are the descriptive two-hour circular block resample, not confidence intervals. PBL height is scheme-defined across CMP08 and is retained only with that caveat.

Metric	CMP07 (X1 − X0)		CMP08 (X2 − X1)	
	Mean	Signs	Mean	Signs
T2 bias (°C)	+0.001	3/1	+0.37	5/0
10 m wind absolute error (m s^{-1})	+0.001	2/2	-0.145	0/5
Surface-pressure bias (Pa)	-0.12	0/4	+6.3	5/0
Stability $S_{50:500}$ (K)	+0.004	3/1	+0.11	3/2
Transport wind $U_{50:500}$ (m s^{-1})	+0.006	4/0	+0.098	5/0
PBL height (m)	+0.01	3/1	-9.2	2/3

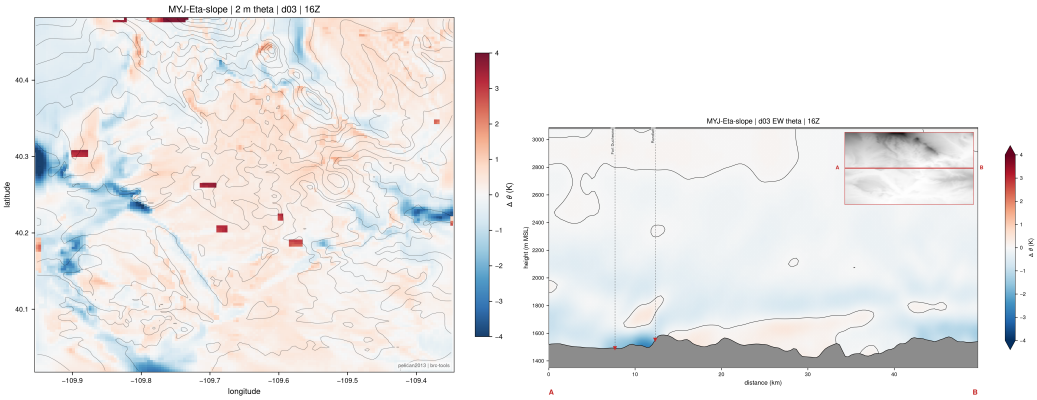


Figure 23. [Exploratory render.] PBL/surface-layer scheme response (CMP08, X2−X1, MYJ/Eta minus YSU/MM5) at 16:00 UTC: 2 m potential temperature (left) and east–west potential-temperature section (right). Batch engine renders behind gate ANA-18.

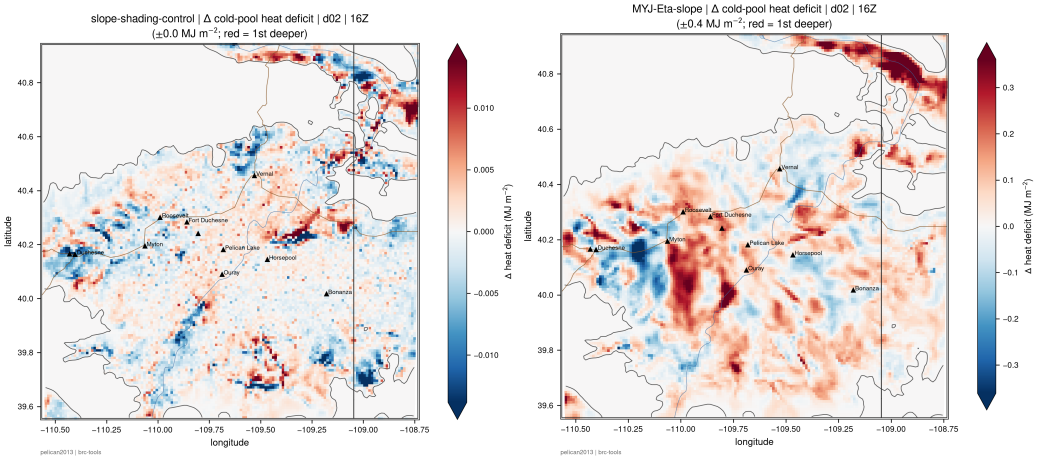


Figure 24. [Exploratory render.] Heat-deficit difference fields at 16:00 UTC for the two process pairs: slope radiation and shading (left, CMP07) and MYJ/Eta scheme (right, CMP08), on adaptive scales. Batch engine renders behind gate ANA-18.

3.11. Initialization Context: Driving-State Profiles

[Diagnostic — not gate-passed: Model-only context. Horsepool has no colocated operational sounding, and the nearest operational radiosondes lie outside the basin, so nothing in this subsection is an observation-model comparison.]

At the 12:00 UTC initialization, the modeled potential-temperature profiles at the Horsepool interior site already separate by driving product below roughly rim level, with the GFS-driven state warmer and less stratified than the NAM-driven state (Figure 25). The corresponding skew-T displays for the two two-way members carry the same lead in conventional coordinates. This initialization separation motivates the wording "driving-product response" used throughout: the products differ from the first output time, before any within-window evolution, and the present design cannot apportion the subsequent response between that inherited state and the lateral boundaries.

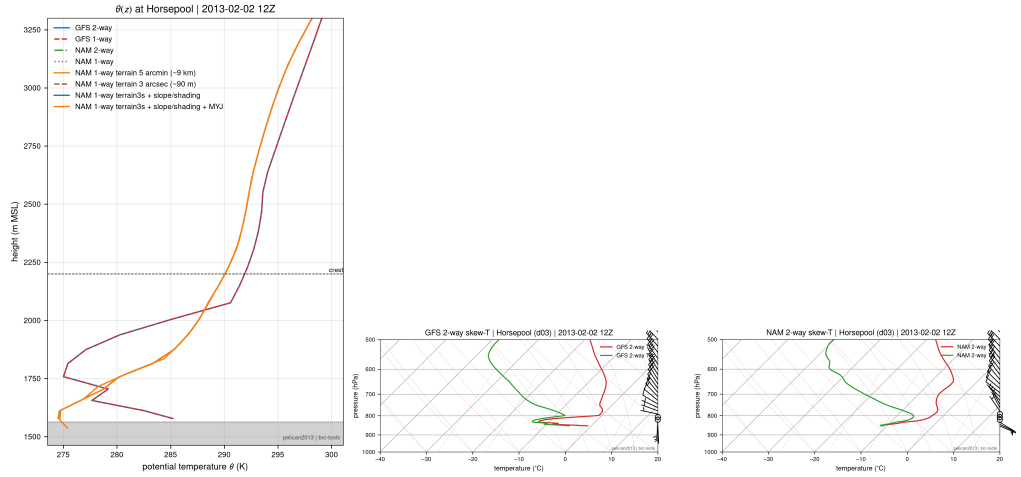


Figure 25. [Exploratory render.] Modeled initialization-time (12:00 UTC) vertical structure at the Horsepool interior site: potential-temperature profiles across the run family (left) and skew-T displays for the GFS two-way (center) and NAM two-way (right) members. Batch engine renders; model-only context with no colocated observed profile.

3.12. What the Archive Cannot Test: the Apparent Resolution Ladder

The archive contains runs whose innermost grids span nominal 1 km, 333 m, and 111 m spacing (X7, X0, and X8), and a visually tempting "resolution ladder" could be drawn from them. That ladder would be false, and this is a confirmed negative design finding (C16)

rather than an omission: the three configurations differ simultaneously in initialization and forecast age, domain count and placement, feedback mode, vertical levels, terrain treatment, radiation options, and output cadence (Table 8). No grid-spacing sensitivity can be attributed within this set, and nominal fine spacing is not evidence of a better cold-pool simulation. A controlled resolution experiment would require a new matched family; the present draft uses X7 and X8 only as design lessons. Legacy nudged and SIP-era simulations in the archive are likewise heterogeneous in episode, model version, and assimilation state, and none enters the frozen result matrix.

Table 8. Why X7/X0/X8 is not a resolution experiment: dimensions on which the members of the apparent ladder differ simultaneously (C16). Any one row would already confound a grid-spacing attribution.

Dimension	Differences across X7 / X0 / X8
Innermost nominal spacing	1 km / 333 m / 111 m
Domain count and placement	different nest counts, extents, and centers
Integration length and age	24 h integration for X7 against 6 h for X0 and X8
Feedback mode	two-way (X7) against one-way (X0, X8)
Vertical levels	75 (X7, X0) against 50 (X8)
Terrain treatment	default-class against 3-arc-second static sources and placements
Radiation options	slope-aware options differ across the set
Output cadence	hourly output windows differ with integration design

4. Discussion

4.1. A Product-Level Difference, Not a Universal Product Ranking

The principal result is a replicated product-level separation. The GFS-driven members are warmer at the evaluated surface sites, less stable across the fixed 50–500 m layer, faster in the fixed-layer transport metric, and lower in footprint-integrated heat deficit than the NAM-driven members under both feedback settings. Replication across the feedback flag makes this ordering insensitive to that one configuration choice within the present experiment. It does not establish that NAM is generally superior, and the opposite ordering of the surface-pressure errors shows that even the present surface evaluation does not support a single-metric ranking. The product factor combines the 12:00 UTC initial atmosphere with boundary updates through 18:00 UTC. A direct comparison of each product with an independent profile at initialization would test inherited lower-tropospheric error, and an experiment that holds either the initial or boundary state fixed would be needed to apportion the subsequent response. Until those tests exist, "driving-product response" is more accurate than "initial-condition effect."

4.2. What the Difference Means for an Air-Quality Model

The GFS members supplied a less stable, faster-moving, and lower-heat-deficit lower layer over the evaluated columns and footprint. That combination is consistent with a meteorological state more favorable to dilution than the matched NAM state. The distinction is relevant because a chemistry calculation is constrained by the mixing volume and transport in its meteorological input; changing chemistry alone cannot test whether a missing shallow accumulation layer caused a concentration error. The footprint heat deficit extends the fixed-column result to an area integral of the simulated pool: the GFS members carried 0.70 to 1.80 MJ m^{-2} less deficit at every retained hour under every footprint definition. However, the transport-wind metric does not measure pollutant export, the heat-deficit footprint is not an observed volume, and no chemical species were simulated. This boundary is important in the Uinta Basin literature. Inventory changes, snow and cloud treatment, deposition, chemical mechanisms, and meteorological confinement have

all produced substantial responses in designs that were often cumulative, bundled, or conditional on one meteorological platform [1,2,4]. An improved meteorological state need not yield improved ozone if another component changes or compensates. The present result therefore identifies a meteorological precondition for attribution rather than assigning an ozone effect or ranking all sources of model error.

4.3. A Predeclared Heat-Deficit Test and Its Negative Result

The heat-deficit analysis carried two criteria that were fixed before any output was inspected. The weaker-state criterion passed unanimously and extends the fixed-column stability result to an area-integrated measure of the simulated pool. The faster-loss criterion failed unanimously because the heat deficit of every member increased through the morning window. We report that outcome rather than relaxing the criterion after inspection. The correct reading is a state offset that was already present at the first retained hour, not a demonstrated within-window divergence in pool evolution between the products. A hypothesis that the GFS members progressively deplete the simulated pool is not supported on this window. Testing differential heat-deficit loss at all would require a longer matched integration through at least one diurnal cycle, with the same predeclared footprint and criteria.

4.4. Where the Driving-Product Response May Live

The executed ANA-19 analysis converts the spatial hypothesis into a split verdict. The confinement half is now demonstrated in committed summaries: the response is concentrated in the 50–500 m layer over the footprint at every pair-hour, with confinement ratios growing from about 1.5 to 3.1 through the window, and the footprint-mean signs match the fixed-column result throughout (C30). The coherence half failed its predeclared bar: at 14:00–15:00 UTC roughly a third of the footprint area carries the opposite 2 m sign, so the gate's two-thirds criterion fails and no spatial figure is promoted, even though the same criterion passes strongly from 16:00 UTC onward. The honest reading is that the driving-product response is real, shallow, and mean-coherent, but not yet spatially unanimous at the start of the retained window under this criterion; a future gate could predeclare a different bar, and this draft does not. The sections add an unanalyzed lead — a secondary elevated warm layer near 2500–3200 m MSL above a rim-level cool band — and the upper-air candidates point the same way, with a more coherent difference at 700 hPa than at 500 hPa. The initialization profiles of Section 3.11 add the temporal end of the same picture: the separation appears to be inherited from the driving state at 12:00 UTC rather than developed within the window. An experiment that swaps initial and boundary sources between the products remains the designed follow-up.

4.5. Heat-Deficit Structure as a Mechanism Question

The footprint heat deficit was introduced as an evaluation integral, but its spatial fields invite a mechanism question: where does the simulated pool hold its energy deficit, and which configuration choices move it? The exploratory fields suggest the driving-product difference is basin-floor-wide while the terrain-source differences concentrate on rims, slopes, and tributaries; the fine-terrain feedback difference is speckle at the ± 0.2 MJ m^{-2} scale. Two locally prototyped displays — a single-point phase portrait and a threshold-defined reservoir anatomy — were deliberately excluded from this draft because both remain threshold-sensitive and unproductized; they are recorded here only as candidate diagnostics for a future gate. Any within-window statement about pool evolution must also respect the negative result of the faster-loss test: on this window the products differ as a state offset, and the archive contains no demonstrated divergence in pool evolution. A defensible

mechanism analysis therefore needs a longer matched integration and predeclared spatial-summary rules before any claim about deficit gain or loss patterns is made.

4.6. Terrain as Geometry and Capacity, Not a Score Shortcut

The terrain bracket is scientifically attractive precisely because its aggregate scores are unremarkable: if terrain-source choice reorganizes the simulated pool along rims and tributaries without moving station means, then station-mean skill is the wrong instrument for terrain decisions in this basin. The executed ANA-17 analysis sharpens that framing in both directions. The fine bracket turns out to be physically small on this grid — an 8 m root-mean-square terrain change producing a 0.1–0.3 K, slope-organized response — and it failed the predeclared terrain-correlation criterion early in the window, so no terrain slot is filled from CMP09. The coarse brackets are the opposite: an order of magnitude larger in both terrain change and response, strongly terrain-correlated at every hour, and now cleanly controlled because the X4 lineage check closed the archive-path caveat (C29). The committed per-station elevation table makes the pressure-artifact mechanism concrete: the coarse member's near-zero mean pressure bias coexists with large station terrain displacement, a reminder that pressure agreement can be a terrain artifact rather than skill. A coarse-bracket geometry gate is therefore the natural next predeclaration; until one passes, no terrain ranking of any kind is offered.

4.7. Process Treatments as Bounded Questions

The two process pairs posed different kinds of questions, and the executed screen answered both. The slope-radiation pair delivered a bounded, instructive outcome: the treatment acts exactly where its physics says it should — an aspect-organized skin-temperature contrast of one to two kelvin on the sidewalls — while remaining invisible to station means on the basin floor and to the archived flat-plane shortwave diagnostic (C31). That is a measurement lesson as much as a physics result: evaluating slope-aware radiation in this archive requires skin-temperature or newly output slope-corrected flux fields, and any future CMP07 gate must be predeclared on instruments the archive can actually express. The scheme pair delivered the coherent result the provisional C19 hinted at: consistently signed shifts across temperature, wind error, pressure, and transport wind (C32), one order below the driving-product contrast, with the instructive internal split that temperature error worsens while wind error improves. The honest framing remains a process-treatment comparison rather than a scheme skill contest, because PBL height is scheme-defined and only the MYJ member exposes `TKE_PBL`. The screen therefore selects CMP08 as the sole Main C6 candidate on coherence and information gain, with its main-versus-supplement disposition left to review. Neither pair alters the paper's anchor result: both are NAM one-way configurations at fine terrain, inside which the driving-product contrast of the factorial is the larger effect by an order of magnitude on every shared diagnostic.

4.8. Nest Feedback as a Replicated Robustness Result

With the CMP06 recomputation (C28), the feedback question now has a consistent answer at two terrain regimes: within-product feedback effects are one to two orders of magnitude smaller than the driving-product contrast at default terrain and at the fine-terrain control, with sensitivity intervals that span zero for the primary lower-layer metrics in both regimes. The only consistently signed fine-terrain feedback effects — slightly larger absolute 10 m wind error and slightly lower surface pressure in the two-way member — are small enough to carry no configuration recommendation on their own. This is the appropriate role for the feedback factor in this study: a replicated robustness check that protects the driving-product conclusion, not a thesis. The result is specific to this case,

window, domain geometry, and diagnostic set, and it licenses no general claim about nest feedback in other regimes or seasons.

4.9. Design Lessons from Confounded and Legacy Configurations

The apparent resolution ladder earns its place in the paper as a refusal. Simultaneous confounds across X7, X0, and X8 (Table 8) mean that any skill difference among them is unattributable, and presenting them as a resolution result would manufacture a conclusion the archive cannot support. The same discipline applies to legacy nudged and SIP-era runs: they motivate design requirements — matched initialization, single-lever differences, recorded as-run namelists, and provenance fingerprints — but they cannot be ranked against the controlled family. These negative design findings are inexpensive to state and materially useful to the next experiment, which is why they are retained in a paper otherwise built from positive controlled contrasts.

4.10. A Queue of Small Supported Findings

Several paragraph-sized findings from the evidence records support the main threads and are collected here with their statuses, so that the submission revision can decide which to keep. Four elevation-spanning stations separate repeatedly from 31 January through 3 February (confirmed, C23), placing the five modeled hours inside an evolving multi-day surface event. The d03 interior domain excludes most bench and basin-end stations while d02 retains the full frozen sample (confirmed, C03), which fixes the evaluation grid choice. Model-minus-station elevation at the eleven fixed columns is median -2.0 m with range -32.4 to +14.3 m (confirmed, C24), so gross column displacement does not drive the factorial result, though representativeness limits remain. Native PBL height does not reproduce the driving-product separation (confirmed, C20), which blocks the most tempting single-variable explanation. The GFS-minus-NAM difference appears more coherent at 700 hPa than at 500 hPa at matched hours (diagnostic), a vertical-context lead now joined by the registered sections' elevated warm layer near 2500–3200 m MSL (diagnostic). The slope-radiation pair's aggregate null coexists with a one-to-two-kelvin aspect-organized skin-temperature response (C31), converting the earlier warning about aggregate metrics into a demonstrated case. The coarse-terrain member combines near-zero mean pressure bias with large station terrain displacement — now directly checkable in the committed per-station elevation table under the closed X4 lineage (C29) — demonstrating that pressure skill and terrain geometry must be separated before pressure is used to rank anything. Each remaining diagnostic item is queued behind a future predeclared gate and none is load-bearing for the paper's conclusions.

4.11. Why the Common-AGL Result Leads the Paper

This draft gives the common-AGL panel priority over both the surface scores and the two-site index. The surface metrics disagree with one another: temperature favors NAM strongly, pressure favors GFS consistently but weakly, and the wind-error ordering depends on the error definition. A single aggregate surface score would therefore conceal exactly the layer behavior this experiment was designed to expose. The two-site index is useful descriptive context and supplies the paper's only direct observation-model comparison, but its stations are horizontally separated, only its two-way product pair is shown, and it cannot validate a free-air inversion. By contrast, the common-AGL calculation applies the same geometric layer, fixed columns, valid times, and aggregation to all four simulations and directly displays both experimental factors. Native PBL height is not a replacement for that calculation: its product contrasts changed sign, and its physical meaning depends on the boundary-layer scheme. This behavior follows the broader lesson from data-assimilation

and cold-pool studies that diagnosed PBL depth, nudging reach, and evaluation-grid structure must be disclosed rather than treated as interchangeable [2,5].

4.12. Limits That Determine the Next Experiment

The five-hour window is the strongest limit. It can show a coherent short-window separation but cannot establish multi-day retention, event buildup, or cold-pool breakup, and the heat-deficit trajectories cover the same five hours. The two-hour spin-up exclusion has not been varied, and resampling five dependent hourly medians cannot create an inferential sample. The observation network creates a second limit. Four stations establish surface context and eleven fixed columns support matched comparisons, but neither set observes the modeled vertical layer, and small nearest-cell elevation errors do not remove exposure and representativeness error. Independent in-basin vertical observations are required to evaluate inversion depth, layer stability, and the simulated heat deficit directly. Observation provenance and the full model-configuration record must also be completed before submission, as marked by the rewrite gates in this draft. All five evidence gates of the selected results architecture have now been executed with predeclared criteria (the complete decision record is the hybrid gate review): the fine-terrain feedback bound passed (Section 3.8); the heat-deficit maintenance leg, the spatial-coherence conjunction, and the fine-terrain-bracket geometry criteria failed and are reported as negatives with their margins; and the process screen selected the boundary-layer-scheme contrast as the sole remaining candidate slot, pending editorial review. Failed gates were not relaxed after inspection, and any renewed spatial or terrain claim requires a freshly predeclared criterion rather than a rescue of a failed one. Only after those steps should selected meteorological states be crossed with emissions and chemical mechanisms in a coupled attribution experiment.

5. Conclusions

Four matched WRF simulations of the 2 February 2013 Uinta Basin episode show a coherent five-hour response to the driving-data product on the common d02 evaluation sample. Relative to NAM, GFS is warmer at the evaluated surface sites, less stable from 50 to 500 m AGL, faster in the fixed-layer transport-wind metric, and 0.70 to 1.80 MJ m^{-2} lower in footprint-integrated heat deficit at every retained hour, under both one-way and two-way nest feedback. Surface-pressure errors order the products in the opposite direction, so aggregate surface skill alone would not identify these lower-layer differences. A predeclared faster-loss criterion for the GFS heat deficit failed in all fourteen pair-definition combinations because the deficit increased in every member; the driving-product response is a state offset within this window, not a demonstrated difference in pool evolution. The within-product feedback responses are much smaller for all evaluated diagnostics, and a matched recomputation at the fine-terrain control pair reproduces that bound, with every fine-terrain feedback effect one to two orders of magnitude below the matched driving-product contrast and with sensitivity intervals spanning zero for the primary lower-layer metrics. All five evidence gates of the selected results architecture were executed for this revision with predeclared criteria: beyond the passing feedback bound, the spatial analysis demonstrated shallow confinement while failing its floor-coherence bar at the first two retained hours, the terrain analysis closed the coarse-member lineage caveat while the fine bracket failed its correlation bar, and the process screen recorded a measurement-limited radiation outcome while selecting a coherent boundary-layer-scheme contrast — warmer temperature bias, smaller wind error, stronger transport wind, all consistently signed — as the one remaining candidate result. Failed gates are reported as negatives with their margins; nothing was promoted, and the surrounding exploratory material sharpens the

questions the next revision can answer without altering the present conclusions. The result establishes a short-window difference in the meteorological state available to downstream chemistry; it does not identify the responsible product component, validate inversion depth, demonstrate event-scale persistence, calculate basin export, or predict ozone. The next discriminating steps are a longer matched factorial with an independent in-basin vertical reference and a design that separates initial from lateral-boundary contributions.

Author Contributions: [Rewrite gate: Assign and approve CRediT roles for all authors.]

Funding: [Rewrite gate: Insert verified funders and grant numbers.]

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The analysis scripts, derived hourly and paired summaries, heat-deficit tables, run and comparison inventories, provenance records, and promoted figures used for this draft are retained in the private working repository. The exploratory renders reproduced in this working revision are catalogued with source paths and status in `figures-draft/HARVEST.md`. **[Rewrite gate: Add a public repository or DOI, licenses and access conditions for raw WRF and Synoptic data, and the exact release commit before submission.]**

Acknowledgments: [Rewrite gate: Identify computational facilities, data providers, and contributors who do not meet the authorship criteria.]

Conflicts of Interest: [Rewrite gate: All authors must confirm the conflicts-of-interest statement.]

Use of Artificial Intelligence: Generative artificial intelligence tools, including Anthropic Claude and OpenAI Codex, were used during manuscript preparation to assist with literature synthesis, provenance and analysis-code review, and drafting and editing. The authors reviewed and edited the resulting material and take responsibility for the content.

References

1. Ahmadov, R.; McKeen, S.; Trainer, M.; Banta, R.; Brewer, A.; Brown, S.; Edwards, P.M.; De Gouw, J.A.; Frost, G.J.; Gilman, J.; et al. Understanding high wintertime ozone pollution events in an oil-and natural gas-producing region of the western US. *Atmos. Chem. Phys.* **2015**, *15*, 411–429.
2. Neemann, E.M.; Crosman, E.T.; Horel, J.D.; Avey, L. Simulations of a cold-air pool associated with elevated wintertime ozone in the Uintah Basin, Utah. *Atmos. Chem. Phys.* **2015**, *15*, 135–151. https://doi.org/10.5194/acp-15-135-2015.
3. Davies, M.J.; Lawson, J.R.; O'Neil, T.; Lyman, S.N.; Zager, K.; Coxson, T.D. Uinta Basin snow shadow: Impact of snow-depth variation on winter ozone formation. *Air (Basel)* **2025**, *3*, 22. https://doi.org/10.3390/air3030022.
4. Matichuk, R.; Tonnesen, G.; Luecken, D.; Gilliam, R.; Napelenok, S.L.; Baker, K.R.; Schwede, D.; Murphy, B.; Helmig, D.; Lyman, S.N.; et al. Evaluation of the Community Multiscale Air Quality Model for Simulating Winter Ozone Formation in the Uinta Basin. *J. Geophys. Res. D: Atmos.* **2017**, *122*, 13545–13572. https://doi.org/10.1002/2017JD027057.
5. Tran, T.; Tran, H.; Mansfield, M.; Lyman, S.; Crosman, E. Four dimensional data assimilation (FDDA) impacts on WRF performance in simulating inversion layer structure and distributions of CMAQ-simulated winter ozone concentrations in Uintah Basin. *Atmos. Environ. (1994)* **2018**, *177*, 75–92. https://doi.org/10.1016/j.atmosenv.2018.01.012.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.